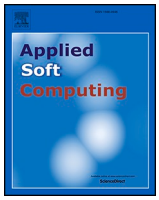
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Criterion-guided polypharmacy side effects prediction with dual-view contrastive learning

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HIGHLIGHTS

- Integration of Molecular Structure and Biochemical Knowledge.
- Application of Contrastive Learning for Drug Embeddings.
- Criterion-Guided Decoder for Polypharmacy Side Effect Prediction.

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ABSTRACT

Predicting side effects from taking multiple drugs (polypharmacy) is a critical challenge in healthcare, with significant implications for patient safety. Traditional machine learning methods often overlook the multi-scale nature of drug information and the interdependencies among side effects, limiting their predictive power. To address these gaps, we propose CASE (Criterion-guided polyphArmacy Side Effects prediction with dual-view contrastive learning). CASE captures both microscopic (molecular graph) and macroscopic (biochemical knowledge) drug features via an adaptive substructure encoder and a biochemical feature aggregator, respectively, under a dual-view framework. Then, contrastive learning is employed to jointly balance and integrate structural and biochemical representations, enabling a more comprehensive and synergistic understanding of drug interactions. Moreover, a criterion-guided decoding strategy is designed to model complex side effect relationships. This effective design enables the proposed CASE model to achieve superior performance, with an accuracy of 88.70 %, precision of 83.62 %, recall of 96.45 %, F1-score of 89.56 %, AUROC of 94.75 %, and AUPRC of 91.16 %, respectively, on the benchmark dataset. Ablation experiments and case studies further confirm the robustness and practical utility of the model. These results demonstrate that CASE can effectively assist clinical decision-making by identifying high-risk drug combinations and delivering reliable polypharmacy risk assessments, thereby supporting safer and more personalized treatment strategies.

1. Introduction

Polypharmacy—the concurrent use of multiple medications to manage coexisting diseases or chronic conditions—has become increasingly common in clinical practice [1]. While it can improve therapeutic efficacy and patient outcomes [2], polypharmacy also introduces serious safety concerns due to complex Drug–Drug Interactions (DDIs), which may trigger adverse side effects. These effects, when unanticipated, can compromise treatment success or even endanger lives. Given the high cost, time constraints, and scalability issues of experimental validation, accurate computational methods for predicting polypharmacy-induced side effects are urgently needed.

To address this need, computational approaches have been widely adopted for predicting polypharmacy-induced side effects [3–5]. In particular, conventional methods typically formulate this task as a multi-relational link prediction problem, which generally involves two main stages: drug representation learning and side effect prediction. The goal of the representation learning stage is to capture the complex

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characteristics of drugs, often achieved by encoding them into latent feature spaces using deep neural networks. For example, Graph Neural Networks (GNNs) have been employed to model drug-drug interaction graphs by iteratively aggregating neighborhood information to produce context-aware drug embeddings [6,7]. Subsequently, the prediction stage aims to uncover potential multi-relational interactions between drug pairs. This is commonly realized through carefully designed decoders that operate within diverse side effect embedding spaces. Among the most widely used techniques in this stage are Multi-Layer Perceptrons (MLPs) and matrix factorization methods, which estimate the likelihood of adverse effects based on the previously learned drug embeddings [8–12]. While many methods have been proposed for polypharmacy side effect prediction, two core motivations drive our work: capturing the multi-scale complexity of drug information, and modeling the nuanced interplay among side effects—both essential for advancing clinical reliability.

While the two-stage paradigm has demonstrated effectiveness in modeling polypharmacy side effects, it often faces challenges in scenarios with limited labeled data or complex, multi-modal drug information. To address these issues, recent studies have begun incorporating self-supervised techniques such as Graph Contrastive Learning (GCL), which enhance representation learning by aligning complementary views of drug entities, without relying on supervision from task-specific labels. This is particularly suited for polypharmacy scenarios, where drugs exhibit heterogeneous biochemical profiles—ranging from chemical structures and protein-binding targets to phenotypic effects—and where aligned, multi-view representations are essential for robust side effect prediction. Several recent studies [13,14] have shown that applying contrastive learning to drug-related prediction tasks improves the quality of drug representations. This improvement stems from encouraging consistency across complementary views, particularly under conditions of limited or noisy supervision.

While recent advances underscore the promise of contrastive learning in drug-related applications, existing methods still lack a unified framework that fully integrates multi-scale biochemical features with side effect modeling. To address this, we revisit polypharmacy side effect prediction through a GCL lens, focusing on two critical challenges: the incomplete representation of drug properties and the overlooked multiplicity and interdependence of side effects.

(1) Incomplete drug representation hinders accurate prediction

While contrastive learning offers a promising paradigm to learn robust representations, current GCL-based DDI models still struggle to capture substructure-level semantics and multi-view biochemical context in a unified and adaptive manner. This issue manifests in three main ways: (i) Lack of comprehensive multi-view drug information. Most existing approaches rely heavily on a single source of information—such as molecular graphs or chemical similarity—and fail to leverage the complementary strengths of other biologically meaningful data sources. In reality, drug behavior under combination therapy is governed by both microscopic structural details (e.g., functional groups, substructures) and macroscopic contextual knowledge (e.g., SMILES fingerprints, protein binding profiles). Ignoring either perspective leads to partial or misleading representations. (ii) Inflexible modeling of internal drug semantics. Drug molecules exhibit significant diversity in terms of structural size, shape, and functional composition, with specific substructures often being directly responsible for triggering adverse effects. However, many existing models apply fixed-depth message passing or pre-defined receptive fields, which lack the capacity to adaptively capture substructures of varying sizes and irregular shapes. Consequently, subtle structural variations that correlate with adverse outcomes may be underrepresented. (iii) Sparse supervision in real-world datasets. Most polypharmacy side effect datasets suffer from limited coverage and class imbalance, making it difficult to learn expressive drug representations from labeled data alone. To address this, contrastive learning enables representation learning from unlabeled data by aligning semantically meaningful views. This allows the model to better capture complex drug characteristics despite limited supervision. Taken together, these limitations underscore the need for a new representation paradigm—one that fuses heterogeneous features, captures substructure-level semantics, and incorporates biochemical context to enhance prediction reliability. To this end, we propose a unified encoding framework that integrates fine-grained molecular structures with complementary biochemical characteristics—such as drug similarity, SMILES fingerprints, and drug–protein associations—as illustrated in Fig. 1(a).

(2) Overlooked Multiplicity and Dependency of Side Effects

In contrastive learning-based models for DDI, significant attention has been devoted to learning rich drug representations. However, the decoder, which determines the emergence of specific side effects, remains underexplored. While many existing studies emphasize building sophisticated encoders to obtain accurate drug embeddings, the decoder—which ultimately determines which side effects emerge from a given drug pair—is often treated as a simple prediction module, such as an inner product or a shallow MLP. This imbalance limits the model's ability to reflect the complex and heterogeneous nature of drug-induced side effects. Importantly, two key characteristics of polypharmacy side effects are frequently overlooked. (i)When a patient takes a combination of two drugs, it often leads to multiple side effects rather than just a single one. (ii)There may be interdependencies between these side effects, whose patterns are drug-pair-specific and influenced by the pharmacological properties of the combination. For example, as shown in Fig. 1**(b)**, when patients concurrently take leucovorin (folinic acid) and Cytosar-U (cytarabine), they may experience multiple adverse reactions, including status epilepticus and aseptic meningitis. Notably, clinical observations have suggested that aseptic meningitis may contribute to or aggravate the onset of status epilepticus. This highlights both the multiplicity of adverse effects and their potential causal relationships. Although some early models, such as Decagon[6] attempt to capture side effect interactions using a global matrix, such static representations cannot adapt to the drug-specific heterogeneity in how these interactions manifest. Relying on a shared interaction matrix for all drug pairs risks missing fine-grained and clinically relevant patterns. Therefore, we argue that an effective decoder should explicitly account for both the multiplicity and dependency of side effects, conditioned on the specific drug pair. Incorporating these aspects into the decoder is crucial for enhancing the accuracy and clinical interpretability of polypharmacy side effect predictions, ultimately contributing to improved drug combination strategies in real-world practice.

To tackle the challenges outlined above, we introduce a novel approach called Criterion-Guided Polypharmacy Side Effects Prediction with Dual-View Contrastive Learning (CASE). Specifically, **for the first challenge,** CASE takes into account both the molecular structure of drugs and the comprehensive biochemical knowledge associated with them, and constructs two distinct views: one based on the molecular structure of drugs (the MS-based view) and another based on biochemical information (the BI-based view). In the MS-based view, CASE focuses on drug substructure that directly affects the chemical properties of drugs and uses an adaptive substructure encoder to capture the chemical structure of drugs in fine granularity. In the BI-based view, CASE considers drug-protein association, drug similarity, and simplified molecular input line entry (SMILE) sequences that describe the global information of drugs. In addition, a biochemical comprehensive feature aggregator is proposed to integrate these three parts of information thoroughly and cohesively. Moreover, drawing inspiration from the successes of contrastive learning, we employ contrastive learning techniques to maximize the mutual information between drug representations generated from the two views. This approach enables the fusion of multi-scale drug information, enhancing the overall robustness and

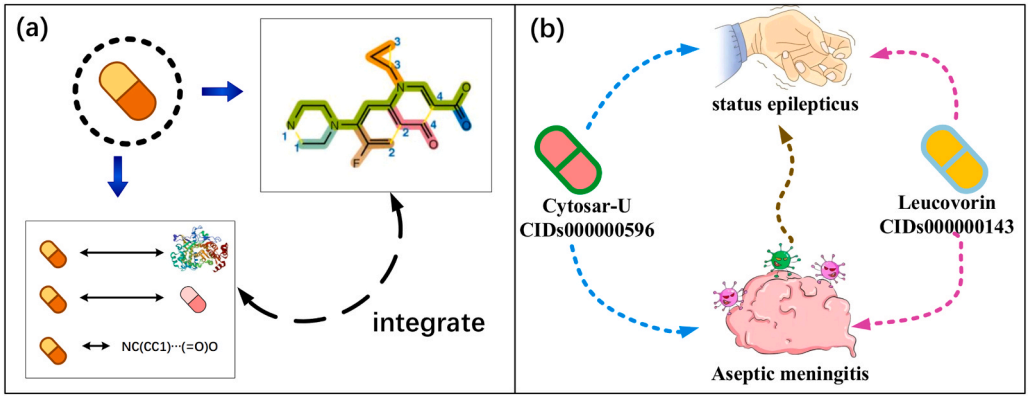


Fig. 1. (a) Accurate drug representation is crucial for predicting downstream side effects. A purely structural view—centered on molecular graphs—offers a chemically grounded foundation, particularly when substructures are carefully captured. However, relying solely on molecular structures often fails to reflect the broader biological and relational context of drugs. To address this limitation, we integrate multiple biochemical perspectives, including drug–target interactions, drug–drug similarity, and SMILES-based fingerprints. This multi-view integration strategy enables a more comprehensive and informative representation of drug entities. (b) An illustrative example of side effect multiplicity and dependency in polypharmacy. When Cytosar-U (cytarabine) and Leucovorin (folinic acid) are co-administered, patients may experience multiple adverse reactions, including status epilepticus and aseptic meningitis. Among them, clinical evidence suggests that aseptic meningitis may act as a contributing factor to status epilepticus.

effectiveness of the model. **For the second challenge**, a criterion-guided strategy is meticulously crafted to account for the potential correlations among different side effects. This strategy enables collaborative training on the probabilities of all predicted side effects, thereby maximizing the exploration of high-quality embeddings produced by the encoder.

In summary, the key contributions of this paper are as follows:

Dual-View Encoder Integration: We have presented an adaptive substructure encoder and a biochemical comprehensive feature aggregator, collectively capturing drugs' nuanced microscopic and macroscopic properties. Contrastive learning is also employed to refine drug embeddings by synthesizing information from different scales, enhancing their quality and representativeness.

Criterion-Guided Decoder Strategy: We have designed a criterion-guided decoder strategy that explicitly considers the potential correlations among various side effects. This strategy involves collaborative training on the probabilities of all predicted side effects, effectively capturing the latent relationships between them and improving the decoder's ability to predict polypharmacy side effects.

Robust Experimental Evaluation: We have conducted an extensive array of experiments using a real-world dataset. Through systematic comparisons with a range of baseline methods, we have demonstrated our CASE model's superior performance and practical utility in accurately predicting and elucidating the complex nature of polypharmacy side effects.

2. Related works

Recently, efforts to enhance polypharmacy side effects prediction have converged around two primary approaches: those based on molecular structure modeling and those centered on feature engineering.

2.1. Molecular structure modeling-based methods

In recent years, significant efforts have been made to address the drug-drug interaction (DDI) prediction problem using the chemical structures of drugs. For example, MHCADDI [15] uses a common attention mechanism between learned substructures of drug pairs to allow each drug to communicate with the other. However, MHCADDI primarily focuses on the molecular structure of drugs and lacks integration of comprehensive biochemical knowledge, which limits its ability to capture multi-scale drug information and explore potential dependencies between side effects, thereby restricting its predictive performance and generalizability. CMPNN-CS [16] treats edges as gates that control the flow of GNNs messaging and thus delineate molecular structures in a learnable manner. However, the gates are computed before message-passing, which means they do not take full advantage of molecular structure information. Furthermore, SSI-DDI [17] explicitly considers substructures as independent entities that operate on a drug's original molecular graph representation, but its rigid substructure definitions ignore the variability in size and shape of functional groups, which are critical for side effect prediction. DSN-DDI [18] introduces a dual-view substructure learning framework to predict potential side effects of drug pairs by learning from both individual drugs ("intra--view") and drug pairs ("inter-view") substructures. CASTER [19] decomposes drug molecules into substructures, smaller substructures, and atoms according to a hierarchy, and efficiently characterizes functional substructures of drugs using a sequence pattern mining module based on the DDI mechanism. However, DSN-DDI's reliance on predefined intra- and inter-view substructures restricts its flexibility, while CASTER's sequence-pattern mining module struggles with irregular substructures, leading to incomplete representations. SA-DDI [20] uses message passing networks to detect substructures of irregular size and shape and simulate substructure-substructure interactions, but it still overlooks the integration of external biochemical knowledge, which is essential for comprehensive drug characterization. Focusing on substructure interaction recently, Tang et al. [21] propose DSIL-DDI, a pluggable substructural interaction module that extracts the substructure interaction patterns of DDIs. However, this approach is limited to structural graph inputs and does not account for modeling the relationships between pharmacological properties and side effects of drugs. To further enhance representation capacity and interpretability, Niu et al. [22] proposed SRR-DDI, which introduced a substructure refinement mechanism based on self-attention, incorporating drug similarity features to improve the expressiveness of substructures and overall model performance. While the method primarily focused on structural aspects, there remains potential for incorporating richer biochemical information. Chen et al. [23] proposed MSKG-DDI, which integrated chemical structure graphs and knowledge graphs to capture multimodal drug information using a fusion neural layer, exploring the complementarity between different data modalities. Although effective at modality-level fusion, the model may not fully capture the granularity of adaptive substructures within drug molecules. Zhong et al. [24] developed MeTDDI, which employed local–global self-attention and co-attention mechanisms to construct motif-based graphs, supporting fine-grained modeling of substructure interactions and providing interpretability through motif visualization.

Its reliance on predefined motifs, however, may affect its flexibility in handling highly diverse drug structures. Sun et al. [25] proposed HDN-DDI, which constructs hierarchical molecular graphs through interpretable substructure decomposition and models substructure interactions using enhanced dual-view representation learning.

To sum up, molecular structure modeling-based methods typically treat drugs as chemical substances comprised of atoms and chemical bonds, representing them as molecular graphs to extract drug characteristics. However, these methods often treat substructures as having a fixed size and do not account for information about drugs' external interactions. Consequently, they lack a mechanism to capture the deep semantics of drugs effectively and are unable to achieve a fine-grained, accurate representation of drugs.

2.2. Feature engineering-based methods

Feature engineering-based methods predict the probability of drug interactions by exploiting different drug-related similarity features, which are typically based on open-source drug-related side information, e.g., chemical substructures, targets, enzymes, or annotations. Early approaches, such as Vilar et al. [26], were limited in scope, relying solely on molecular structure similarity and ignoring other essential biological information. After 2017, a great deal of work has thrived on fusing the multi-source heterogeneous features of drugs. For example, the Tiresias [27] integrates various sources of drug-related features and knowledge. Then, the final side effect interactions are predicted based on the similarity metrics calculated between the two drugs. However, these metrics are typically handcrafted or empirically defined, and fail to capture the deep and nonlinear relationships inherent in drug-biological interactions. Ma et al. [28] proposed a method that learns drug's embedding representations from multiple types of drug features and adds thoughtful mechanisms to determine the weights of each view, but their model lacks a principled mechanism to ensure the views are complementary. Deng et al. [29] propose an integrative framework DDIMDL that extracts 65 DDI events from DrugBank by events trimming and using the integrated multiple types of drug features to build a prediction mode, but overlooking the interdependencies that are crucial for realistic polypharmacy modeling. Recently, Yang et al. [30] propose CNN-Siam, a convolutional neural network algorithm to learn the feature representation of drug pairs from multimodal data of drugs including chemical substructures, targets, and enzymes. However, this approach simply concatenates features from different modalities without explicitly modeling their complementarities, resulting in suboptimal fusion. Masumshah et al. [31] proposed a polypharmacy side effects prediction framework called DPSP, which uses deep neural networks to predict DDIs. DPSP evaluates various drug information, measures drug similarity through feature extraction and Jaccard similarity, and generates a novel feature vector for each drug to predict specific DDI events. However, DPSP relies heavily on handcrafted similarity measures, which are inherently limited in capturing complex and nonlinear relationships between drugs. Moreover, the framework lacks an adaptive mechanism to weigh the relevance of different feature types and does not explicitly consider interdependencies among side effects. To further enhance the modeling of multi-source drug features, Jiang et al. [32] proposed a hierarchical multi-relational graph representation learning framework (HMGRL), which constructed heterogeneous graphs based on multiple drug similarity sources and learned both explicit and implicit drug-pair correlations through relational graph convolution and multi-view spectral clustering. Asfand-e-yar et al. [33] developed MMCNN-DDI, a multi-modal convolutional neural network framework that incorporated four parallel CNN modules to learn feature representations from drug chemical structures, enzymes, pathways, and targets, and then concatenated them to predict interaction-associated events. Although the model achieved competitive performance, its ability to effectively coordinate information from multiple modalities could be further enhanced. Kumari et al. [34] introduced SNF-HCNN, a hybrid similarity-based model that computed similarity matrices from seven drug features, fused them via similarity network fusion, and fed the combined matrix into a hybrid CNN and logistic regression framework. However, its reliance on handcrafted similarity metrics may affect scalability in more complex settings. Li et al. [35] proposed DABI-DDI, a dual-attention and Bayesian inference-based framework that integrated multi-source drug features with interpretable modeling components. While this method enhanced both interpretability and robustness, its effectiveness may vary depending on the consistency and quality of the input features.

In summary, feature engineering modelling-based methods place excessive dependence on intricate feature engineering and frequently necessitate substantial expert domain knowledge. A notable limitation of these approaches is their inability to function when features and domain knowledge about drugs are unavailable. Furthermore, many existing models tend to utilize simple deep neural networks for final predictions without adequately analysing the potential impact of considering side effects, which may result in suboptimal performance. To offer a comprehensive overview of mainstream approaches in polypharmacy side effect prediction, we summarize and compare different categories of models mentioned above, as shown in Table 1.

3. Problem formulation

In this section, we will explain the definition of the polypharmacy side effect prediction task. In order to facilitate the introduction of the definition, we introduce several basic concepts. For readability, we summarize the main notations in CASE, as shown in Table 2. Notations. We denote tensors by uppercase boldface letters (e.g. $\mathbf{X}$), vectors by lowercase boldface letters (e.g. $\mathbf{x}$), normal font for scalars (e.g. x), and calligraphic font for sets (e.g. $\mathscr{X}$).

Definition 1. (polypharmacy side effects prediction). The polypharmacy side effect prediction task can be modelled as a multi-label classification problem. In this problem, we define the set of drugs as $\mathscr{D} = \{d_1, d_2, \ldots, d_{|\mathscr{D}|}\}$, whose corresponding set of molecular structure graph is denoted as $\mathscr{G} = \{G_1, G_2, \ldots, G_{|\mathscr{D}|}\}$. $\mathcal{R} = \{r_1, r_2, \ldots, r_{|\mathcal{R}|}\}$ is defined as the set of side effects. We also define a side effect events tensor $\mathbf{Y} \in \mathbb{R}^{|\mathscr{D}|\times|\mathscr{D}|\times|\mathcal{R}|}$. The polypharmacy side effect prediction task takes as input two drugs d_i and d_j, and outputs predicted values $\hat{y}_{ij}^k$ for interaction r_k that may exist between them. Our goal is to learn a prediction function $\hat{y}_{ij}^k = F((d_i, r_k, d_j)|\Theta, \mathscr{D}, \mathscr{G}, \mathbf{Y})$, where Θ denotes the model parameters.

Definition 2. (drug molecular graph). Each drug molecule graph can be considered a graph with atoms as nodes and bonds as edges. $\mathscr{G}$ is defined as the set of molecular graphs. Given $G_i \in \mathscr{G}$, G_i is denoted by $G_i = (\mathscr{V}_i, \mathscr{E}_i)$, where $\mathscr{V}_i$ represents the set of nodes, $\mathscr{E}_i$ represents the set of edges, $v_{i,n} \in \mathscr{V}_i$ denotes the n-th atom and the bond connecting the n-th and m-th atoms is denoted by $e_{i,nm} \in \mathscr{E}_i$.

4. Methodology

Before introducing CASE in detail, we describe its architecture in Fig. 2. Clearly, the prediction of polypharmacy side effects using CASE comprises two crucial components: i) **Dual-View Contrastive Learning Encoder**: This component involves an adaptive substructure encoder and a biochemical comprehensive feature aggregator in both the MS-based view and BI-based view. In the MS-based view, a messaging network with a substructure attention mechanism is proposed to encode the drug molecular graph structure into a low-dimensional embedding vector that adaptively captures the substructure. In the BI-based view, CASE integrates comprehensive drug biochemical knowledge, which consists of three components that capture different characteristics of drugs. Besides, the proposed drug representation contrastive learning method is employed to refine the drug representations generated under both views. ii) **Criterion-Guided Decoder:** This component takes into

Table 1
Comparative summary of representative DDI prediction methods across molecular and feature engineering paradigms.

Category	Method	Strengths	Limitations
Molecular structure modelling-based methods	MHCADDI	Uses multi-head cross-attention to model substructure-level interactions.	Lacks biochemical integration
	CMPNN-CS	Uses edge-gated message passing to learn molecular graph representations.	Gates computed before message passing
	SSI-DDI	Treats substructures as independent, improving fine-grained molecular modeling	Fixed substructure shapes
	DSN-DDI	Introduces intra-view and inter-view substructure learning across drug and drug-pair levels.	Rigid intra-/inter-view definition
	CASTER	Decomposes drug molecules into hierarchical structures for detailed substructure mining.	Struggles with irregular subgraphs
	SA-DDI	Identifies irregular-shaped substructures and models substructure-substructure interactions.	Ignores external biochemical data
	DSIL-DDI	Provides a pluggable module to capture fine-grained substructure interaction patterns.	Misses pharmacological-side effect links
Feature engineering-based methods	Tiresias	Integrates heterogeneous biomedical features (e.g., ATC codes, enzymes, targets).	Handcrafted metrics
	AttTransGAE	Learns drug embeddings across multiple feature views with adaptive weights.	No complementarity guarantee
	DDIMDL	Leverages DrugBank-curated DDI events and multiple drug properties for prediction.	Ignores inter-feature dependency
	CNN-Siam	Uses convolutional siamese networks to learn multimodal drug pair representations.	No explicit modality fusion modeling
	DPSP	Combines similarity metrics and deep neural networks for DDI prediction.	Relies on fixed similarity, lacks adaptivity

Table 2
Main notations in CASE. The top rows are for dual-view contrastive learning; the bottom rows cover the criterion-guided decoder and training loss.

Notation	Description
$\mathscr{G}$	Set of molecular graphs.
$\mathscr{V}_i, \mathscr{E}_i$	Atom set and bond set of the i-th molecular graph.
$\mathbf{h}_{i,nm}$	Directed representation of bond from atom $v_{i,n}$ to $v_{i,m}$of drug d_i.
$\mathbf{h}_{i,m}$	Atom representation for the i-th atom of drug d_i.
α_i	Atom-centered attention score of drug d_i.
$\mathbf{a}_i$	Graph-level representation of drug d_i based on atoms.
$\mathbf{h}_i$	Final graph-level representation in MS-based view of drug d_i.
$\mathbf{A}^p, \mathbf{A}^d, \mathbf{A}^s$	Drug-protein association matrix, drug similarity matrix and drug SMILE fingerprint matrix.
$\mathbf{e}_i$	Representation in the BI-based view of drug d_i.
$\hat{y}_{ij}$	General overall prediction of drug pair(d_i, d_j).
$\hat{\mathbf{Y}}$	General overall prediction matrix.
$\hat{y}_{ij}^k$	Probability of side effects r_k between drug pairs (d_i, d_j)
$\mathbf{Q}$	Drug heterogeneity matrix.
$\mathbf{U}^k, \mathbf{B}^k$	Upper bound matrix and the bottom bound matrix under the k-th side effect.
$\mathscr{L}_{CG}^k, \mathscr{L}_{CL}, \mathscr{L}$	Hinge loss under the k-th side effect, the contrastive loss and the total loss.

account different criteria of implied heterogeneous data by introducing polypharmacy-specific boundaries, to consider the potential impact of other side effects when predicting the probability of target side effects.

4.1. The MS-based view

In this subsection, we construct the MS-based view, which focuses on the molecular structure of drugs, and employ an adaptive substructure encoder to probe into the substructures that affect the chemical properties of drugs. Specifically, adaptive substructure coding encompasses three stages: *directed message passing*, *atom-centered attention aggregation*, and *substructure aware pooling*. Among them, the latter two are designed based on chemical intuition, that is, polypharmacy side effects can be caused by chemical substructures of different sizes and shapes. By adaptive substructure coding, the substructures can be captured and characterized, affecting the chemical properties of drugs. This fine-grained representation mechanism is helpful to enhance the accuracy of drug molecule representation, and thus provide a reliable basis for the prediction of polypharmacy side effects.

4.1.1. Directed message passing

The representation of chemical bonds between atoms directly impacts the representation of molecules. However, the message-passing network commonly used in chemistry makes it challenging to pass atomic messages between nodes in a clear direction, resulting in difficulties in accurately capturing global and local information about molecular structure. To effectively capture the structural information of drug molecules and minimize unnecessary circulation in the information transmission trajectory, we utilize directional message passing to generate fine-grained chemical bonds in the molecular graph G_i of drug d_i to represent $\mathbf{h}_{i,nm}$:

$$\mathbf{h}_{i,nm}^{(l+1)} = \mathbf{h}_{i,nm}^{(0)} + \sum\nolimits_{v_k \in \mathscr{N}(v_n) \setminus v_m} \mathbf{h}_{i,kn}^{(l)} \tag{1}$$

where $\mathbf{h}_{i,nm}^{(l+1)} \in \mathbb{R}^t$ denotes directed representation of bond from atom node v_n to node v_m at layer $(l+1)$. Note that $\mathbf{h}_{i,nm}^{(l+1)}$ is different from $\mathbf{h}_{i,mn}^{(l+1)}$. $\mathscr{N}(v_n)$ is the neighbourhood set of atom node v_n. Following Yang et al.'s work [20], we obtain the initial directed representation of bond $\mathbf{h}_{i,nm}^{(0)} \in \mathbb{R}^t$ and the initial representation of atom $\mathbf{h}_{i,m}^{(0)} \in \mathbb{R}^{t'}$. Specifically, the initial dimension of bond representations is denoted as t, and the initial dimension of atom representations is denoted as t'. These notations are consistently used in the subsequent formulas.

4.1.2. Atom-centered attention aggregation

Traditional aggregation methods often depend on fixed-size molecular substructures, potentially resulting in incomplete representation of drug information. To address this limitation, we introduce an atom-centered attention mechanism. As illustrated in Fig. 3, this mechanism assigns adaptive importance scores to substructures with varying radii around each atom, allowing the model to dynamically focus on the most informative substructures that are potentially responsible for adverse drug reactions. By assigning higher attention to key substructures, the specific substructures directly linked to side effects are highlighted during the aggregation process. This enhancement significantly boosts the interpretability and precision of CASE when predicting and analyzing drug side effects. Specifically, we first aggregate the chemical bond representation of each atomic node to get the representation of each node, and then further aggregate these node representations to obtain the graph-level representation $\mathbf{c}_i^{(l)} \in \mathbb{R}^t$ at the l layer:

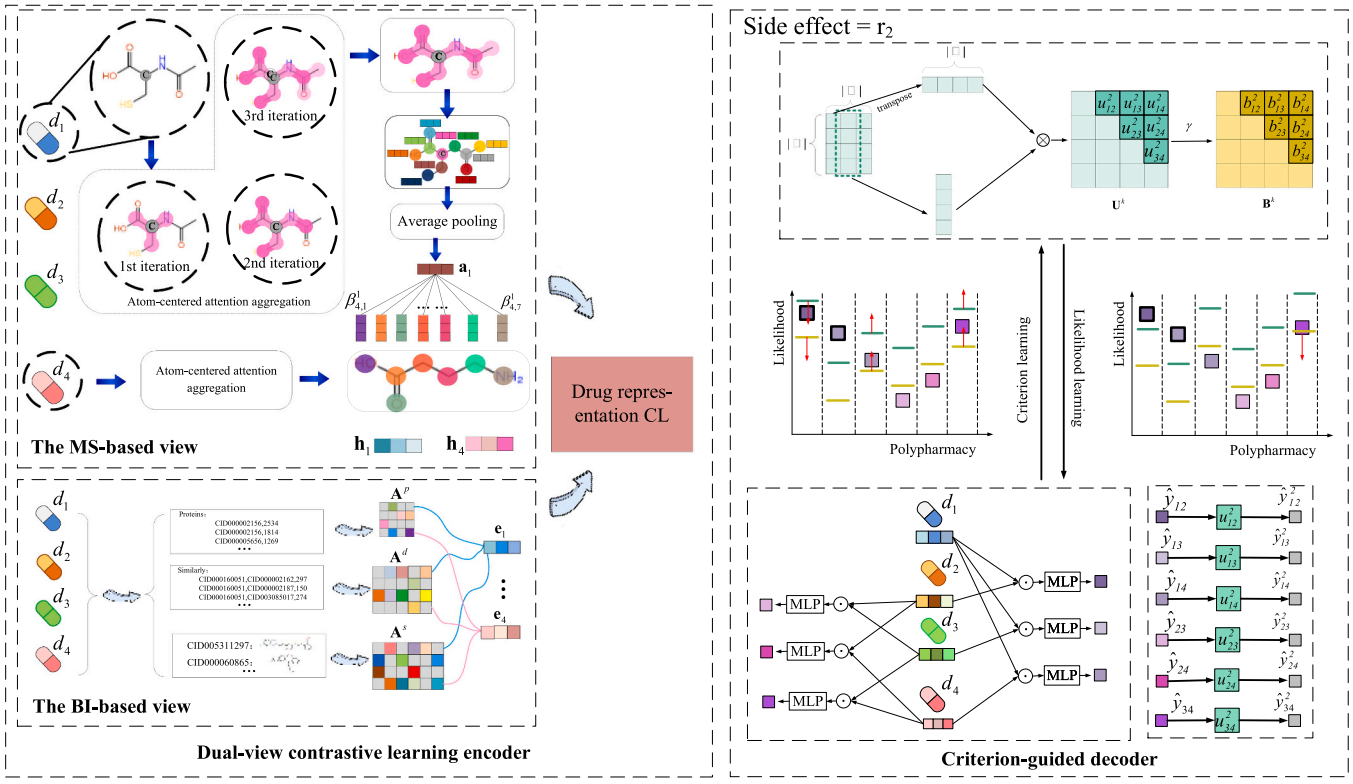


Fig. 2. The overall architecture of CASE. Component-1: Dual-view contrastive learning encoder. MS-based view estimates the molecular structure of drugs through adaptive substructure encoder, and BI-based view extracts the biochemical comprehensive knowledge of drugs through biochemical feature aggregator. Contrastive learning of drug representation is used to refine the drug embeddings generated in two views. Component-2: Criterion-guided decoder. The drug representations of the two views are fused and fed into the criterion-guided decoder, which will first obtain the overall initial prediction of all drug pairs under all side effects, then obtain the upper bound $\mathbf{u}^{(k)}$ and the lower bound $\mathbf{b}^{(k)}$ under the k-th side effect according to the initial prediction, and finally be trained by the upper and lower bound guidance model. Among them, the small red arrow indicates the training process of the criteria. Note: Fig. 2. utilizes four drugs (d_1, d_2, d_3, and d_4) as illustrative examples to showcase the model's process. In the MS-based view, the depicted steps demonstrate how molecular structure-based representations of d_1 and d_4 ($\mathbf{h}_1$ and $\mathbf{h}_4$, respectively) are obtained. It is important to highlight that the calculation process for any other pair of drugs follows the consistent methodology. In the BI-based view, all four drugs are chosen to extract biochemical characteristics, providing a more vivid explanation of $\mathbf{A}^{\mathrm{p}}$, $\mathbf{A}^{\mathrm{d}}$ and $\mathbf{A}^{\mathrm{s}}$ as matrices.

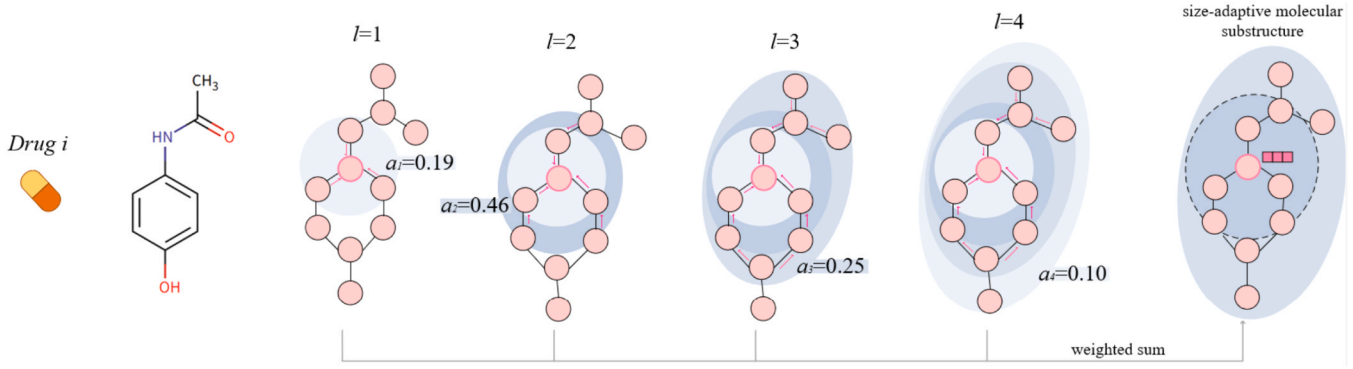


Fig. 3. Illustration of the atom-centered attention aggregation mechanism. The model adaptively assigns attention scores to local substructures with different radii centered around each atom, enabling the prioritization of side-effect-relevant substructures during graph-level representation learning.

$$\mathbf{c}_i^{(l)} = \sum_{m=1}^{|\mathcal{V}_i|} \left(\frac{1}{|\mathcal{N}(v_m)|} \sum_{v_n \in \mathcal{N}(v_m)} \mathbf{h}_{i,nm}^{(l)} \right) \tag{2}$$

where $\mathcal{N}(v_m)$ is the neighbourhood set of atom node v_m. Eqs. (1) and (2) produce graph-level representations that successively incorporate larger radii. Consequently, these graph-level representations, which consider different radius sizes, can capture molecular substructures of different scales. To identify the drug substructure that directly affects the occurrence of side effects and to convey this substructure information to the central atomic node of the substructure, we use MLP to map $\mathbf{c}_i^{(l)}$ into scalar $c_i^{(l)}$. Then, the Softmax function is utilized to normalize scores of different iteration layers to derive the attention score $\alpha_i^{(l)}$, as calculated below:

$$\alpha_i^{(l)} = \frac{\exp(c_i^{(l)})}{\sum_{k=1}^{L} \exp(c_i^{(k)})} \tag{3}$$

This aggregation approach enables the assignment of different levels

of importance to various substructures, ensuring that information of the appropriate scale is captured. This facilitates the representation of specific molecular substructures at the atomic node level $\mathbf{h}_{i,m} \in \mathbb{R}^t$ as follows:

$$\mathbf{h}_{i,m} = ReLU\left(\mathbf{W}_1\left(\mathbf{W}_2\mathbf{h}_{i,m}^{(0)} + \sum_{v_n \in \mathcal{N}(v_m)} \sum_{l=1}^{L} \alpha_i^{(l)} \mathbf{h}_{i,nm}^{(l)}\right) + \mathbf{b}_1\right) \tag{4}$$

where $\mathbf{W}_1 \in \mathbb{R}^{t \times t}$ and $\mathbf{W}_2 \in \mathbb{R}^{t \times t'}$ are the weight matrices, $\mathbf{b}_1$ is the bias, and ReLU(·) denotes the Rectified Linear Unit activation function.

4.1.3. Substructure aware pooling

Through atom-centered attention aggregation, atomic representations are derived that encapsulate the neighbourhood information, capturing the significant structural features. In line with the principles of medicinal chemistry [36], a drug consists of various functional groups and chemical substructures. These components are integral in defining the pharmacokinetic properties of a drug as well as its pharmacodynamic effects. Therefore, to further distinguish key substructures from trivial substructures in the representation of aggregated atoms, we suppose that d_j 's structural characteristics can disclose d_i's potential internal substructure, and we advance substructure-aware pooling. Employing the probability assessment of side effects emerging from the interaction between drugs (d_i, d_j) as an example for analysis, we depict the process of deriving the molecular graph-level representation of drugs by utilizing atomic representations that contain vital structural neighbourhood information.

Specifically, the possibility of potential interaction is calculated by using the n-th atom representation $\mathbf{h}_{i,n}$ of drug d_i (which contains important structural neighborhood information) and the graph-level representation $\mathbf{a}_j \in \mathbb{R}^t$ of drug d_j. In addition, an interaction probability score $\beta_{i,n}^j$ is assigned to the n-th atom of d_i. The calculation process is as follows:

$$\mathbf{a}_j = \frac{1}{|\mathcal{V}_j|} \sum_{m=1}^{|\mathcal{V}_j|} \mathbf{h}_{j,m} \tag{5}$$

$$\beta_{i,n}^j = \mathrm{Softmax}\left(\left(\mathbf{W}_i\mathbf{h}_{i,n}\right)^\top \left(\mathbf{W}_j\mathbf{a}_j\right)\right) \tag{6}$$

where $\mathbf{W}_i \in \mathbb{R}^{t \times t}$ and $\mathbf{W}_j \in \mathbb{R}^{t \times t}$ are the weight matrices that map the atom and graph representations into the same feature space. Finally, the graph-level representation of d_i can be calculated by:

$$\mathbf{h}_i = \sum_{n=1}^{|\mathcal{V}_i|} \beta_{i,n}^j \mathbf{h}_{i,n} \tag{7}$$

where $\beta_{i,n}^j$ denotes the attention weight assigned to the n-th atom in drug d_i, reflecting its relative importance in drug–drug interactions. By performing a weighted summation over all atom-level representations $\mathbf{h}_{i,n}$, the model adaptively focuses on structurally salient regions that are most predictive of adverse side effects. This attention-guided aggregation mechanism effectively mitigates the information dilution issue inherent in traditional average-based molecular representations, thereby enabling more accurate identification of potential side effects in polypharmacy settings.

4.2. The BI-based view

In this subsection, to thoroughly integrate the multi-scale information of drugs, we construct a view based on biochemical information, i.e. BI-based view. We then apply a biochemical comprehensive feature aggregator to this view, which aims to harness the biochemical knowledge of drugs, including drug-protein interaction matrices, drug similarity matrices, and drug SMILES fingerprint matrices. We begin by elucidating the rationale for selecting this specific knowledge and proceed to discuss how to process the knowledge data to integrate it effectively with the CASE model. Then, we introduce the biochemical comprehensive feature aggregator which can fully integrate drug characteristics.

4.2.1. Drug biochemical comprehensive knowledge

The drug-protein association matrix. Drug–protein interactions represent a fundamental biochemical perspective in polypharmacy side effect prediction. Unlike structural views that focus solely on molecular substructure, this view captures the functional interface between drugs and the biological system, which is indispensable for understanding drug mechanisms at the system level. Specifically, the drug–protein association matrix models the direct interactions between drugs and their known protein targets. These interactions are crucial because proteins often serve as the primary effectors of drug action. Drug-induced perturbations to protein activity can cascade into downstream effects, such as gene expression changes, signalling disruption, or metabolic imbalance, that ultimately contribute to adverse side effects. [37].

This perspective is complementary and non-redundant to other biochemical inputs. For example, while chemical similarity may indicate functional resemblance between drugs, it cannot capture the specific biological consequences mediated by protein-level interactions, which often underlie unanticipated side effects. Likewise, SMILES sequences encode the chemical structure of a drug but lack direct biological context. In contrast, the drug–protein view embeds drugs within the biological interaction network, enabling the model to reason about biochemical causality and functional outcomes.

To operationalize this drug-protein network perspective, we utilize data sourced from a seminal study focused on side effect prediction [6]. This data set integrates various drug and protein networks. To express the drug-protein association, a drug-protein matrix is constructed, which is denoted as $\mathbf{A}^p \in \mathbb{R}^{|\mathcal{D}| \times p}$, where p is the number of proteins. Suppose that a drug interacts with some protein, we assign a value of 1 at the corresponding position in the matrix and a value of 0 at all other positions.

The drug similarity matrix. Drug similarity plays a pivotal role in characterizing the structural, chemical, and functional relationships between compounds. It provides a relational perspective that complements individual drug features, enabling the model to reason over pairwise interactions. However, obtaining complete side effect data in laboratory or clinical trials can be expensive, time-consuming, and challenging. To address this challenge, utilizing drug similarity allows extending known side effect information to similar drug pairs, thereby enhancing the understanding of polypharmacy side effects. As such, it offers a practical approach for leveraging existing knowledge to support inference in data-sparse regions. This contributes to improved safety assessment, drug development, and clinical decision-making.

Unlike input features that capture only intrinsic properties of individual drugs—such as molecular structure or protein binding profiles—drug similarity inherently embodies a pairwise structure, making it well-suited for modeling relational patterns that underlie drug–drug interactions. Additionally, similarity scores provided by databases such as STITCH integrate diverse biomedical evidence, including chemical structures, pathway co-membership, literature co-occurrence, and experimental associations. This multimodal aggregation renders the similarity matrix not merely a handcrafted relational prior, but a compact, data-driven representation of pharmacological context. Its incorporation into the model thus enriches interaction modeling and supports the inference of complex side effect mechanisms with greater biological fidelity.

We quantify the similarity between drugs by employing the interaction scores derived from the STITCH database [38]. Specifically, we initiate by performing compound ID mapping between the data of multi-drug side effects and STITCH. Subsequently, we select 'combined_score' as the similarity metric between two drugs. Given that the

combined_score values span from 1 to 999, we normalize the drug similarity values to lie within the range of 0–1 to facilitate meaningful interpretation. Finally, we construct a drug similarity matrix $\mathbf{A}^d \in \mathbb{R}^{|\mathscr{D}|\times|\mathscr{D}|}$.

The drug SMILE fingerprint matrix. The Simplified Molecular Input Line Entry System (SMILES) encodes a drug's molecular structure as a linear string that sequentially represents atoms, bonds, and substructures. This compact representation captures both local and global chemical features in a format amenable to efficient processing by machine learning models. SMILES-based features play a crucial role in side effect prediction, as they capture structural motifs—such as functional groups, ring systems, and metabolic sites—commonly associated with adverse drug reactions. By embedding SMILES strings using models such as Mol2vec [26], it becomes possible to extract context-aware structural information that reflects the co-occurrence of substructures in chemically meaningful patterns. This facilitates the identification of subtle chemical signatures associated with toxicity or drug–drug interactions, even in the absence of explicit annotation.

Within the multi-source information fusion framework proposed in this study, SMILES fingerprints exhibit distinct necessity and relative advantages. Unlike protein–drug interaction data or drug similarity matrices, which depend on external biological annotations, SMILES-based features provide a stable and self-contained molecular representation grounded solely in chemical composition. This makes them a critical complement to other biochemical information sources. Moreover, SMILES representations are widely available for nearly all drug molecules and follow a standardized format, ensuring high accessibility and consistency across samples. This universality facilitates coherent feature alignment in large-scale modeling, thereby improving model generalization.

We use STITCH as our primary data acquisition source to retrieve the SMILE sequence and match drugs in the polypharmacy side effect data set with their respective compound IDs. Subsequently, SMILE sequences are fed into the Mol2vec model to obtain the SMILE embedding of drugs [39]. To represent each drug with its corresponding embedding, we construct a matrix $\mathbf{A}^s \in \mathbb{R}^{|\mathscr{D}|\times s}$, where s represents the embedding dimension.

4.2.2. Biochemical comprehensive feature aggregator

The final representation of drugs in the BI-based view is obtained by concatenating the drug-protein association feature, drug similarity feature, and drug SMILE fingerprint feature. These concatenated features are then input into a MLP to extract the low-dimensional representation of drugs. Taking drug d_i as an example, the drug representation $\mathbf{e}_i \in \mathbb{R}^t$ in BI-based view can be calculated as follows:

$$\mathbf{e}_i = \mathrm{Re}LU\left(\left(\mathbf{A}^p_{i:}||\mathbf{A}^d_{i:}||\mathbf{A}^s_{i:}\right)\mathbf{W}_3 + \mathbf{b}_3\right) \tag{8}$$

where $\mathbf{W}_3 \in \mathbb{R}^{(p+|\mathscr{D}|+s)\times t}$ is the weight matric, and $\mathbf{b}_3$ is the bias. The symbol "||" denotes matrix concatenation.

In this way, we can comprehensively use drug-protein association, drug similarity and drug SMILE fingerprint features to describe the characteristics of drugs. This helps provide more informative input for CASE, thus improving the performance and reliability of the polypharmacy side effect prediction task.

4.3. Drug representation CL

To achieve a more comprehensive and effective drug embedding, we have developed both microscopic and macroscopic perspectives for drugs, yielding corresponding representations. Specifically, in the MS-based view, we employ an adaptive substructure encoder to capture the molecular structure characteristics of drugs. In the BI-based view, the biochemical information of drugs is synthesized from various types of macro-interactive data. As a result, the drug embedding representations learned from these two views are not only relevant but also mutually complementary. They mutually guide each other's training, indicating a collaborative trend. Therefore, understanding the mechanism from the above heterogeneous cross-view comparison becomes crucial for the model's effectiveness. This aligns with the hypothesis of the contrastive learning framework. Motivated by this perspective, we use InfoNCE as the objective function of contrastive learning [40]. For each drug d_i, the calculation process of contrastive learning is as follows:

$$\mathscr{L}_{CL} = -\log\sigma(f_D(\mathbf{e}_i, \mathbf{h}_i)) - \log\sigma(1 - f_D(\widetilde{\mathbf{e}}_i, \mathbf{h}_i)) \tag{9}$$

where $\widetilde{\mathbf{e}}_\mathbf{i}$ is negative sample obtained by corrupting $\mathbf{e}_\mathbf{i}$ with row-wise, and the discriminator $f_D(\cdot)$ is defined as the dot product between vectors. The Sigmoid function is chosen as the activation functionσ. Through this calculation process, contrastive learning is introduced to enhance the characterization ability of drugs, better capture multiple characteristics of drugs, and maintain consistency between heterogeneous views.

4.4. Criterion-guided decoder

Current polypharmacy side effect prediction models typically rely on nonlinear neural networks to assess if a drug pair will lead to adverse effects. However, the above models have the following limitations. Firstly, methods such as MIRACLE [9] have struggled with the task of distinguishing between specific types of side effects. Merely identifying the presence of side effects is insufficient to provide clinically informative guidance. Secondly, traditional approaches often fall short in exploring the nuanced relationships among different side effects, leading to an overdependence on the encoder for prediction results. Thus, there is a clear need for more sophisticated strategies that can effectively capture and dissect the complex interactions between side effects to improve the precision and dependability of predictions. To deal with these issues, we propose a novel criterion-guided decoder in this study, which trains all the side effects to be predicted by drugs collaboratively, resulting in improved prediction accuracy. The criterion-guided decoder enhances the accuracy of polypharmacy side effect prediction, providing more valuable guidance for treatment decision-making. Specifically, for each drug pair (d_i, d_j), we initially derive their general prediction $\widehat{y}_{ij}$:

$$\widehat{y}_{ij} = \sigma\left(\mathbf{W}_4\left(\mathbf{h}_i \odot \mathbf{h}_j\right) + \mathbf{b}_4\right) \tag{10}$$

where $\odot$ denotes the element-wise product. $\mathbf{W}_4$ is the weight matric, and $\mathbf{b}_4$ is the bias. It is worth noting that $\widehat{y}_{ij}$ is the general prediction of (d_i, d_j), encapsulating the potential for a general interaction between the two drugs. Next, we decode $\widehat{y}_{ij}$ under different side effects.

For each drug pair (d_i, d_j), we obtain the general prediction and construct the predictions of all drug pairs as a general prediction matrix $\widehat{\mathbf{Y}} \in \mathbb{R}^{|\mathscr{D}|\times|\mathscr{D}|}$. Since our purpose is to predict polypharmacy side effects instead of single-drug side effects, and side effects are independent of the order of the drugs, the predicted values of drugs d_j and d_i are equivalent to those of drugs d_j and d_i. Therefore, we ignore the lower triangular part of the matrix. The upper triangular matrix excluding the diagonal is expanded by rows and defined as a general overall prediction vector $\widehat{\mathbf{y}} \in \mathbb{R}^{1\times N}$ where $N = |\mathscr{D}|(|\mathscr{D}| - 1)/2$. This process is shown in Fig. 4(a).

As shown in Fig. 4(b), we introduce a trainable drug heterogeneity matrix $\mathbf{Q} \in \mathbb{R}^{|\mathscr{D}|\times|\mathscr{R}|}$, where each row corresponds to a drug and each column corresponds to a side effect. Each element in matrix $\mathbf{Q}$ encodes the intrinsic susceptibility or inducing tendency of drug d_i to cause side effect r_k. Biologically, this matrix is grounded in the concept of differential drug-specific sensitivity: drugs often vary in their propensity to induce specific adverse events, depending on their molecular targets, pharmacokinetics, and metabolic pathways. Instead of treating side effects as globally independent events, $\mathbf{Q}$ allows the model to learn side effect-specific response profiles for each drug. To further leverage this property, we decompose $\mathbf{Q}$ column-wise to analyze the risk profile associated with

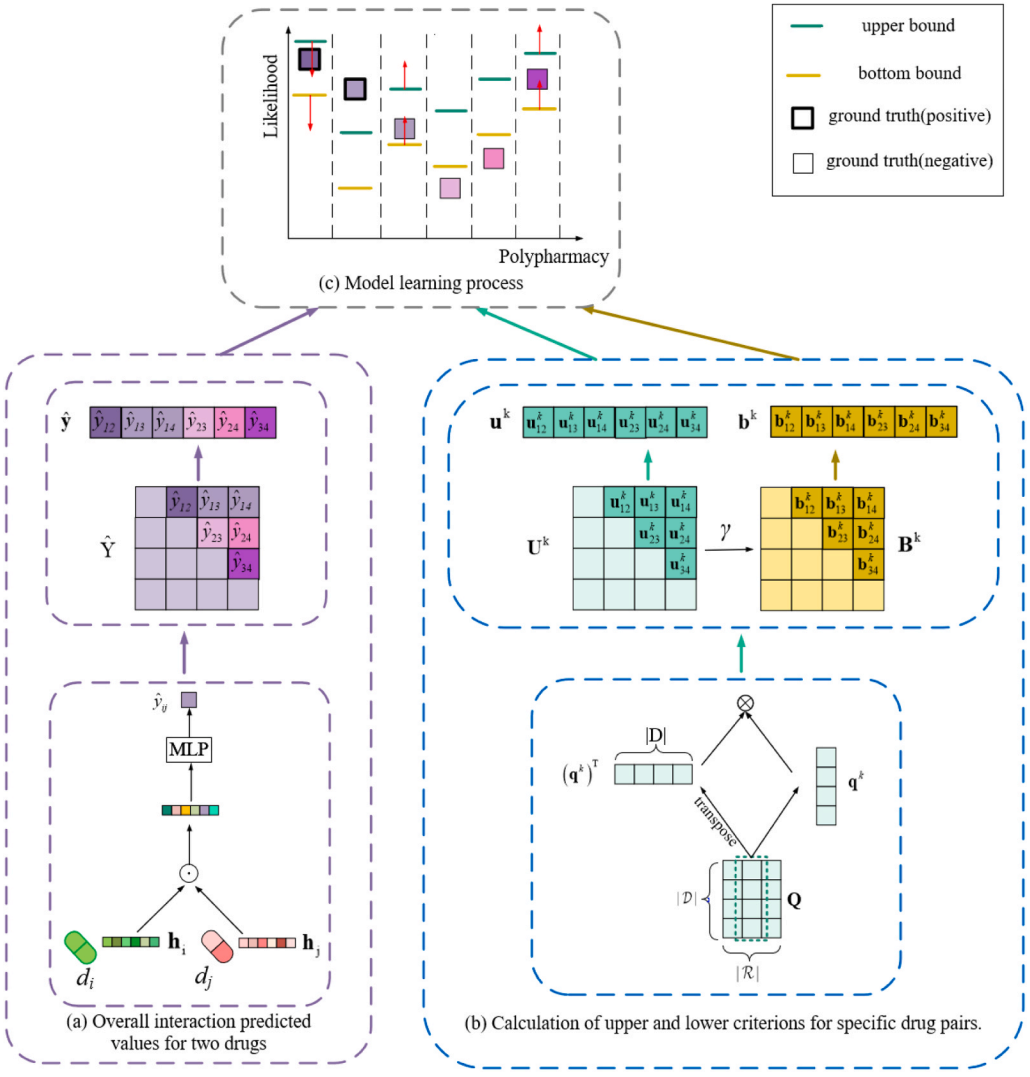


Fig. 4. Computation and learning of upper bounds, bottom bounds and overall prediction values in encoders.

each side effect individually. Specifically, the k-th column of $\mathbf{Q}$, denoted as $\mathbf{q}^k \in \mathbb{R}^{|\mathscr{D}|\times 1}$, represents the drug-wise risk spectrum for side effect r_k—that is, how each drug is predisposed to inducing this particular side effect. We then compute the outer product $\mathbf{q}^k \cdot (\mathbf{q}^k)^{\mathrm{T}}$ to obtain a pairwise risk interaction matrix under this side effect. This matrix models the hypothesis that if both drugs in a pair have high individual propensities toward a side effect, then co-administration amplifies the risk—a well-documented pattern in clinical pharmacology. For example, combining two drugs that both prolong the ventricular repolarization interval (e.g., azithromycin and fluoxetine) significantly raises the likelihood of cardiac arrhythmia, consistent with the additive or synergistic effects modeled by the outer product. Formally, the upper bound matrix $\mathbf{U}^k$ and the lower bound matrix $\mathbf{B}^k$ under the k-th side effect can be calculated as:

$$\mathbf{U}^k = \mathbf{q}^k \cdot (\mathbf{q}^k)^{\mathrm{T}} \tag{11}$$

$$\mathbf{B}^k = \gamma \mathbf{U}^k \tag{12}$$

where $\gamma \in (0,1)$ is a hyperparameter that controls the ratio of the two bounds. The upper bound matrix $\mathbf{U}^k$ represents the outer product of the vector $\mathbf{q}^k \in \mathbb{R}^{|\mathscr{D}|\times 1}$ obtained from the k-th column of the Q matrix and itself. The matrix $\mathbf{U}^k$provides information about the correlation and importance of all drug pairs with the k-th side effect. Each element in the matrix represents the strength of the relationship between a specific drug pair and the k-th side effect, with higher values indicating a stronger correlation. Then, we take the upper triangle part of upper bound matrix $\mathbf{U}^k$ and the lower bound matrix $\mathbf{B}^k$, which does not include the diagonal line, and expand it by rows to get the upper bound vector $\mathbf{u}^k \in \mathbb{R}^{1\times N}$ and lower bound vectors $\mathbf{b}^k \in \mathbb{R}^{1\times N}$ where $N = |\mathscr{D}|(|\mathscr{D}| - 1)/2$.

Fig. 4(c) illustrates the model's training process using hinge loss. The horizontal axis represents different drug pairs based on the type of side effects, while the vertical axis represents the upper/lower bounds along with the corresponding probability values. The purple square in the diagram represents the vector $\hat{\mathbf{y}}$. The blue upper bound and the yellow lower bound correspond to vectors $\mathbf{u}^k$ and $\mathbf{b}^k$, respectively. These vectors are derived from the randomly initialized matrix $\mathbf{Q}$. The red arrow represents the learning process. The loss function penalizes cases where the likelihood is below the upper bound for positive interactions and above the lower bound for negative interactions. To minimize the loss, we jointly update the parameters of the encoder and the parameters of the matrix $\mathbf{Q}$. The hinge loss under the k-th side effect can be defined as:

$$\mathscr{L}_{CG}^k = \sum_{d_i \in \mathscr{D}} \left(\sum_{d_j \in \mathscr{D}_+^{i,k}} \max\left(u_{ij}^k - \hat{y}_{ij}, 0\right) + \delta \sum_{d_j \in \mathscr{D}_-^{i,k}} \max\left(\hat{y}_{ij} - b_{ij}^k, 0\right) \right) \tag{13}$$

where $\mathscr{D}_{+}^{i,k}$ ($\mathscr{D}_{-}^{i,k}$) shows the set of medicines that interact with d_i that cause side effects k or do not cause it. Moreover, u_{ij}^{k} (b_{ij}^{k}) denotes an element in the vector $\mathbf{u}^{k}$ ($\mathbf{b}^{k}$), which represents the upper (bottom) bound of the interaction probability between drug d_i and drug d_j under side effect k. The weight of the negative entry is denoted by δ. We utilize the $\max(\cdot, 0)$ function to compute the upper and bottom bounds for each drug pair under the k-th side effect. Consequently, the loss function only penalizes situations where the probability of a positive interaction is below the upper bound and the probability of a negative interaction exceeds the lower bound.

Finally, the general prediction $\widehat{y}_{ij}$:of a drug pair under all side effects is assigned to each side effect, to obtain the probability $\widehat{y}_{ij}^{k}$ of all side effects occurring for all drug pairs:

$$\widehat{y}_{ij}^{k} = \frac{\widehat{y}_{ij}}{u_{ij}^{k}} \tag{14}$$

This side-effect-specific prediction framework is not merely a mathematical operation but also biologically motivated. In real-world clinical settings, multiple side effects often occur together and are not mutually independent. For example, certain inflammatory reactions may elevate the risk of neurological or metabolic symptoms. In some cases, one adverse effect may act as a trigger or risk factor for another, forming complex conditional pathways. By assigning a general prediction score to each side effect and constraining it using adaptive thresholds, our model implicitly captures such interdependencies. This allows CASE to reflect not only the independent likelihood of each side effect but also their potential co-occurrence, mutual reinforcement, or causal linkage, which are critical for realistic polypharmacy risk assessment.

Hyperparameter ϵ is utilized to add contrastive learning loss $\mathscr{L}_{CL}$ and hinge loss $\mathscr{L}_{CG}^{k}$ to get the total loss $\mathscr{L}$. Furthermore, the whole learning procedure is summarized in Algorithm 1.

$$\mathscr{L} = \sum_{k=1}^{|\mathcal{R}|} \mathscr{L}_{CG}^{k} + \epsilon \cdot \mathscr{L}_{CL} \tag{15}$$

Algorithm 1. The training process of CASE.

Input: Side effect events tensor $\mathbf{Y}$, drug molecule graph set $\mathbf{G}$, drug-protein association matrix $\mathbf{A}^{p}$, drug similarity matrix $\mathbf{A}^{d}$, drug SMILE fingerprint matrix $\mathbf{A}^{s}$, maximum training epochs T.

Output: $F((d_i, r_k, d_j) \mid \Theta, \mathbf{D}, \mathbf{G}, \mathbf{Y}) : \hat{y}_{ij}^{k} \rightarrow [0,1]$.

Initialize Θ with Xavier initialization, in which Θ represents the parameters of the CASE model;
$t \leftarrow 1$;
repeat
  for $d_i \in \mathrm{D}$ **do**
    // adaptive substructure encoder
    for $l \in L$ **do**
      Calculate directed representation of bond $\boldsymbol{h}_{i,nm}^{l}$ using directed message passing via Eq. (1);
      Calculate graph-level representation $\mathbf{c}_i^{(l)}$ based on chemical bonds vie Eq. (2);
    end for
    Calculate atomic node representation of specific molecular substructure $\mathbf{h}_{i,m}$ via Eq. (3) and Eq. (4);
    Calculate graph-level representation $\mathbf{h}_i$ in MS-based view using substructure aware pooling via Eq. (5)(6)(7);
    // biochemical comprehensive feature aggregator
    Calculate $\mathbf{e}_j$ in BI-based view using biochemical comprehensive feature aggregator via Eq. (8);
  end for
  // criterion-guided decoder
  Calculate general overall prediction $\hat{y}_{ij}$ via Eq. (10);
  Calculate upper bound matrix $\mathbf{U}^{k}$ and bottom bound matrix $\mathbf{B}^{k}$ via Eq. (11) and Eq. (12);
  Predict the probability $\hat{y}_{ij}^{k}$ of the existence of the k-th side effect of drug pair (d_i, d_j);
  Calculate drug representation contrastive learning L_{CL} via Eq. (9);
  Calculate hinge loss L_{CG}^{k} via Eq. (13);
  Calculate total loss L via Eq. (15);
  Update Θ by gradient descent;
until $t > T$ or Eq. (16) is converged;
return $F((d_i, r_k, d_j) \mid \Theta, \mathrm{D}, \mathrm{G}, \mathbf{Y})$

4.5. Time complexity

To provide a detailed understanding of the overall computational cost, we perform a step-by-step analysis of the time complexity associated with each module of the proposed CASE model. Specifically, CASE comprises four key components: (1) the MS-based view, (2) the BI-based encoder, (3) the contrastive learning module, and (4) the criterion-guided decoder. We analyse the time complexity of each component individually and identify the dominant computational terms that

determine the model's scalability and efficiency in large-scale polypharmacy side effect prediction tasks.

First, the MS-based view consists of three stages: directed message passing and atom-centered attention aggregation. In the directed message passing stage, the model updates each edge representation once and applies a linear transformation of dimension t. Assuming each drug molecule graph contains on average C edges and there are $|\mathscr{D}|$drugs in total, the overall time complexity for this step is$O(|\mathscr{D}|\cdot C\cdot t)$. Subsequently, the atom-centered attention aggregation mechanism performs a weighted summation over each node's neighboring representations. As this operation also relies on the number of edges C, its computational procedure is analogous to that of the message passing stage and incurs a similar level of computational cost. After obtaining the drug representations, CASE introduces a substructure-aware pooling module to capture structural interactions between drug pairs. The computational complexity of this operation for each drug pair is $O(t^2)$, and given L drug pairs, the overall complexity is $O(L\cdot t^2)$.

Second, The BI-based encoder is designed to integrate multiple biochemical information of drugs, including drug–protein associations, drug similarity, and SMILES fingerprints. These three types of features are concatenated to form a unified representation with dimension $U = (p + |\mathscr{D}| + s)$, which is then projected into a t-dimensional embedding via a multi-layer perceptron. The computational cost of this mapping for each drug is $O(U\cdot t)$, and given $|\mathscr{D}|$ drugs, the total complexity is $O(|\mathscr{D}|\cdot U\cdot t)$.

The contrastive learning module enhances the discriminative capability of representation learning by performing dot product computations and loss evaluations on positive and negative drug pairs. For each pair, two vector dot products are computed, resulting in a computational cost of $O(2t)$. For L drug pairs, the overall computational cost is $O(Lt)$.

Finally, the criterion-guided decoder first calculates the prediction score for each drug pair through a dot product operation, with a time complexity of $O(Lt)$. It then generates the upper and lower bound embeddings for each side effect type using the drug heterogeneity matrix $\mathbf{Q}$. For each side effect, this involves performing an outer product operation between drug embeddings, incurring a computational cost of $O(|\mathscr{D}|^2)$. Given $|\mathscr{R}|$ side effect types, the total computational cost is $O(|\mathscr{R}||\mathscr{D}|^2)$.

In summary, the overall time complexity of the CASE model can be expressed as$O(|\mathscr{D}|\cdot C\cdot t + |\mathscr{D}|\cdot U\cdot t + L\cdot t^2 + |\mathscr{R}||\mathscr{D}|^2)$. Among these terms, $L\cdot t^2$ and $|\mathscr{R}||\mathscr{D}|^2$ represent the primary computational costs associated with large-scale drug pair modeling and side effect dependency modeling, respectively. These two components largely determine the scalability and computational bottlenecks of CASE in real-world applications.

5. Experiments

5.1. Dataset description

We use the following datasets, including polypharmacy side effect data, drug-protein association data, drug similarity data, and drug SMILE fingerprint data:

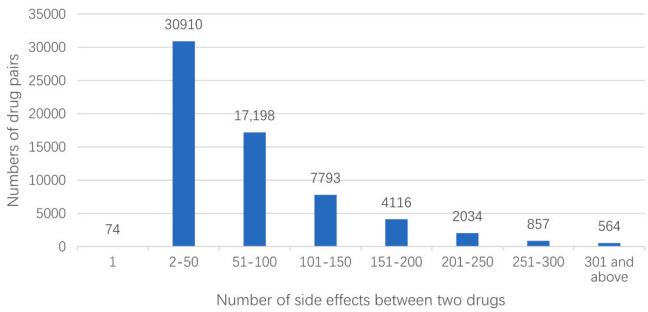


Fig. 5. Distribution of the number of side effects between two drugs.

- *Polypharmacy side effect data & Drug molecular graph.* Following a recent approach [20], we utilize a dataset constructed by filtering and pre-processing the original TWOSIDES[41] database. It is important to highlight that the TWOSIDES is chosen for its comprehensive data on potential side effects between drug pairs. According to the statistics presented in Fig. 5, our analysis identified a total of 63,546 drug pairs that may exhibit one or more side effects. Interestingly, among these pairs, only 74 exclusively manifest a single side effect. This indicates that the percentage of drug pairs with only one side effect is approximately 0.1 %. In comparison, other multi-drug side effects databases like DrugBank[42] contain 191,878 drug-drug interaction (DDI) entries. Among these entries, only 406 drug pairs (0.2 %) are associated with two types of DDI, while the vast majority of drug pairs (99.8 %) are linked to a singular type of DDI. The TWOSIDES database provides data on multiple side effects that may occur between two drugs. This conforms to our purpose for researching the side effects of polypharmacy. In addition, each drug can be converted into a molecular map using RDKit [43].

  While TWOSIDES offers broad coverage of polypharmacy side effects, we acknowledge that it is constructed by mining adverse event reports, and thus may contain noisy or spurious associations. Previous studies (e.g., Yang et al., 2022) have pointed out that TWOSIDES is less suitable for cold-start evaluation due to potential false positives. Nevertheless, it remains widely adopted in multi-label DDI prediction tasks, especially in standard (hot-start) settings. Compared to DrugBank, which primarily focuses on biochemical or mechanistic drug interactions, TWOSIDES captures clinically observable outcomes and includes significantly more drug pairs with multiple side effects. This aligns more directly with our modeling objective: to predict a wide range of clinically relevant adverse reactions that arise when drugs are co-administered. In future work, we plan to validate our model on additional high-confidence datasets such as DrugBank and SIDER, and explore denoising techniques to further improve robustness.
- *Drug-protein association data.* We adopt publicly available drug-protein data from a typical research Decagon. [6]. To obtain the proteins associated with each drug, the list of all drugs in the polypharmacy side effect data matches this data. It is worth noting that among the 645 drugs matched with Decagon's publicly available drug-protein data, only 285 drugs are associated with corresponding target information. This means that the remaining 360 drugs do not have any associated proteins in the dataset. As a result, in the matrix $\mathbf{A}^p \in \mathbb{R}^{|\mathscr{D}|\times p}$, the vectors corresponding to these 360 drugs are set to 0 to indicate the absence of target information for those drugs.
- *Drug similarity & Drug SMILE fingerprint data.* We obtained the drug similarity data set from STITCH to evaluate the similarity between two drugs. "combined_scores" is chosen as a comprehensive indicator because it is a comprehensive metric used to indicate the degree of similarity between chemicals. This score is derived by incorporating various information sources such as experimental data, literature reports, and calculation methods. A higher "combined_scores" suggests a greater similarity. Then, the index is normalized to attain the final drug similarity. Meanwhile, the STITC is chosen as the drug SMILE fingerprint sequences provider.

Table 3
Statistics of datasets. Data Source indicates the databases that store different kinds of data. Data description indicates the size of each type of data.

Data	Data source	Data description
$\mathbf{Y}$	TWOSIDES	645 drugs, 963 side effect types and 4576,287 side effects.
$\mathbf{A}^p$	Decagon	285 drugs and 18,691 drug-protein associations.
$\mathbf{A}^d$	STITCH	56,241 similar drug pairs and 359,784 dissimilar drug pairs.
$\mathbf{A}^s$	STITCH	645 drug SMILE fingerprints.

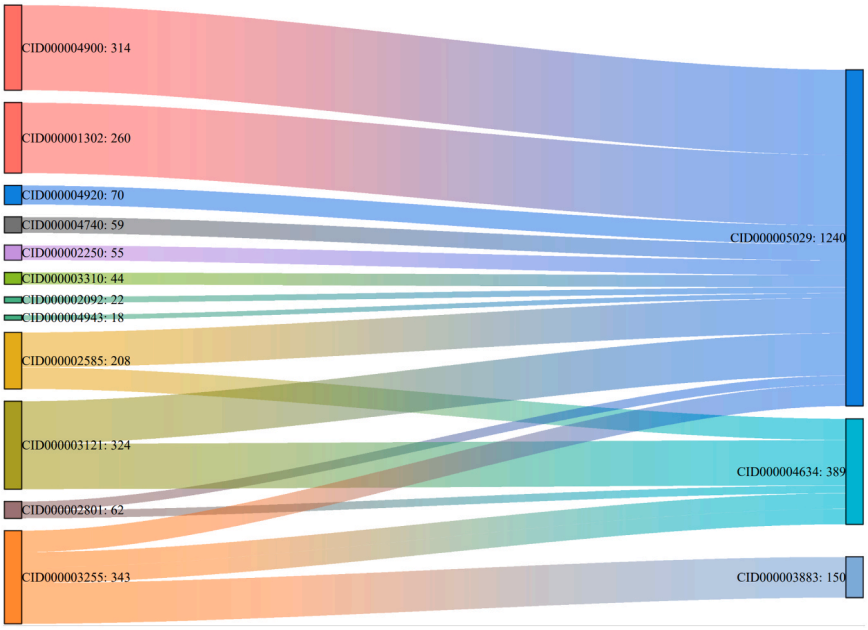


Fig. 6. Examples of polypharmacy adverse reactions.

Table 3 lists the statistical information of the experimental data. To quantify the co-occurrence between side effects, we visualize the number of drug combinations in which side effects co-occur with other side effects in Fig. 5. Due to space constraints, only the intervals of the number of side effects are displayed. Fig. 5. shows that only 74 drug pairs produce a single side effect when taken simultaneously. In contrast, side effects do not occur independently in a huge number of co-prescribed drug pairs. Additionally, we randomly select 17 drug pairs from the polypharmacy data and visualize the number of side effects for these drug pairs in Fig. 6. This shows more clearly that multiple side effects can occur when two drugs are taken together, and that one drug may have side effects with different drugs.

5.2. Experimental settings

5.2.1. Baselines

To assess the superiority of our proposed model, we select the following baseline approaches to compare performance. These methods encompass both traditional and recently published state-of-the-art methods.

- *DeepCCI* [44]: This method applies convolutional neural networks composed of convolutional and pooling layers to capture the specificity of SMILE strings.
- *MR-GNN* [45]: The method learns drug node features using a multi-resolution-based architecture. In addition, it uses bipartite graph state long short-term memory networks to summarize local features of each graph and extract interaction features between pairs of graphs.
- *SSI-DDI* [17]: The method can generate raw molecular graph representations of drugs. Additionally, it treats substructures as independent entities and uses the interactions between substructures as a basis for whether two drugs have potential adverse effects.
- *GMPNN-CS* [16]: This method uses gated messaging neural network to learn drug substructures of different sizes and shapes based on the interaction scores between drug substructures, thus generating fine-grained drug embedding and predicting the probability of side effects between drugs.
- *GAT-DDI* [46]: The GNNs in GMPNN-CS are replaced by graph attention networks to learn the chemical structure of drugs, then the resulting embeddings are used directly for polypharmacy side effect prediction.
- *SA-DDI* [20]: This method can detect drug substructures with irregular size and shape, simulate substructure-substructure interaction by assigning different attention to each substructure, and finally predict the probability of polypharmacy side effects.

5.2.2. Evaluation metrics

To quantify the model's performance, we follow the strategy proposed by [20,47], and the experimental results are presented in the six widely used metrics: Accuracy (ACC), Precision (Prec), Recall (Rec), F1-score (F1), Area Under the Receiver Operating Characteristic (AUROC) and Area Under the Precision-Recall Curve (AUPRC).

5.2.3. Implementation details

In subsequent performance experiments, the CASE model utilizes the same Polypharmacy side effect data and Drug molecular graph data as all baseline methods. The positive samples are directly obtained from known DDI recorded of TWOSIDES, while negative samples are generated following Wang et al.'s strategy [48], where each positive sample is paired with one negative sample (1:1 ratio). This setup allows for a standardized comparison across baseline methods. However, in real-world clinical settings, drug–drug interactions are highly imbalanced, with negative cases far outnumbering positive ones. To address this issue, we further evaluate our model under more realistic imbalanced conditions with negative-to-positive ratios of 1:2 and 1:3. We implemented the CASE framework using PyTorch [49]. For the general setting, the batch size is set to 256, the hidden size to 64, and the number of message-passing iterations (layers) in the graph neural network to 10. The learning rate is fixed at 0.001, and model parameters are updated using the Adam optimizer. The epoch size is set to 200 for all baselines. To make the model have the best performance, coefficients γ and δ of the objective function are 0.5 and 0.1 respectively.

5.3. Experimental results

5.3.1. Overall performance

To ensure a standardized and fair comparison between our CASE model and the baseline methods, we initially followed the conventional data division paradigm [17,50] employed in the baseline methods.

Table 4
Performance (mean ± std in %) of our model CASE against competitive approaches.

	ACC	Prec	Rec	F1	AUROC	AUPRC
DeepCCI	75.16 ± 0.30	71.65 ± 0.68	83.31 ± 0.84	77.03 ± 0.05	82.42 ± 0.31	79.47 ± 0.35
MR-GNN	85.39 ± 0.31	80.57 ± 0.37	93.28 ± 0.21	86.46 ± 0.27	91.93 ± 0.20	89.32 ± 0.22
SSI-DDI	82.21 ± 0.41	79.10 ± 0.31	87.56 ± 0.81	83.11 ± 0.44	89.27 ± 0.38	86.19 ± 0.41
GAT-DDI	67.32 ± 2.04	71.54 ± 2.31	57.62 ± 5.09	63.70 ± 3.28	75.16 ± 2.47	72.50 ± 2.45
GMPNN-CS	86.96 ± 0.03	82.20 ± 0.03	94.35 ± 0.10	87.85 ± 0.04	92.94 ± 0.02	90.38 ± 0.04
SA-DDI	87.45 ± 0.03	82.43 ± 0.02	95.18 ± 0.10	88.35 ± 0.04	93.17 ± 0.04	90.51 ± 0.08
CASE	**88.70 ± 0.09**	**83.62 ± 0.03**	**96.45 ± 0.13**	**89.56 ± 0.08**	**94.75 ± 0.05**	**91.16 ± 0.07**
***p*-value**	**3.41 ×e−07**	**5.66 ×e−06**	**4.52 ×e−05**	**2.38 ×e−05**	**5.42 ×e−06**	**2.53 ×e−05**

Specifically, we randomly split the dataset into three parts: training (60 %), validation (20 %), and test (20 %) sets, which enables a comprehensive evaluation and comparison. The experimental results are presented in Table 4, with the optimal results highlighted in bold. Compared with all the baselines, our approach obtained more favorable outcomes. To assess the statistical significance of the observed improvements, we performed paired t-tests across five independent runs. A *p*-value less than 0.05 indicates that the performance gains of our model, CASE, over the strongest baseline SA-DDI are statistically significant, thereby supporting the robustness and superiority of our approach. All the results reported in this subsection are based on a balanced 1:1 sampling setup, where each positive drug pair is matched with one negative pair. This follows the standard evaluation protocol used in previous studies [20,48]. In addition, to better reflect the imbalanced nature of real-world drug interaction data, we further assess CASE under more realistic negative-to-positive ratios (1:2 and 1:3). These results, discussed in the next subsection, demonstrate the model's robustness in handling skewed data distributions. We have the following observations:

(1) Drug SMILE fingerprint sequences alone are insufficient for obtaining high-quality drug embeddings.

The DeepCCI model shows relatively less comparability among the baseline methods, as it solely depends on drug SMILE sequences while ignoring other information, such as the topological details of drugs. Although SMILEs are frequently employed and are relatively uncomplicated information for depicting drugs, they are beset with limitations, such as the loss of information and the restriction of localized information to a certain extent. To achieve a more accurate representation of the drug, a structure-based approach might be more fitting for tasks related to the side effects of polypharmacy.

(2) The drug embeddings cannot be wholly manifested just based on the chemical structure.

Within the baselines, MR-GNN represents chemical structures as graphs and uses dual graph-state long short-term memory networks (LSTMs) to summarize local features, SSI-DDI operates directly on the raw molecular graph representations of drugs, and GMPNN-CS infers whether a pair of drugs can cause a DDI based on their chemical substructures. These three methods work better than DeepCCI but are not as good as SA-DDI and CASE. Such methods could take into account the topographic features of drug subgraphs, highlighting the importance of incorporating the information regarding the topology of drug structure in the prediction of side effects. Nevertheless, these methods, which depend solely on chemical structure and utilize only a single viewpoint, frequently lead to less-than-optimal modeling performance. GAT-DDI exhibits relatively suboptimal performance, possibly due to gradient vanishing/exploding or smoothing problems.

(3) Elementary encoders and decoders cannot fully encompass the complex information about drugs and the potential associations among side effects.

Although we previously attributed the DeepCCI model's less comparable due to its reliance solely on drug SMILES representation, it's crucial to recognize that this might not be the sole factor contributing to its underperformance. DeepCCI model extracts local patterns from structured data like images using CNNs, but this approach may not be the most appropriate architecture for sequential data such as SMILES strings. MR-GNN model processes structural features at multiple resolutions and employs dual graph-state LSTMs to extract interaction features. However, it may encounter challenges in effectively integrating contextual information from the entire molecular graph. As for GMPNN-CS, it considers edges as gates to control the flow of message passing and delimit substructures. This simplistic approach to edge modeling fails to effectively capture complex interactions involving multiple drugs and physiological pathways, thereby resulting in limited predictive performance. Besides, both SSI-DDI and GAT-DDI rely on Graph Attention Networks (GATs) to obtain representations of molecules. However, both methods are limited by the local receptive field, which hinders the ability of the model to capture the global features of drug molecular structure effectively. SA-DDI has achieved the best performance among all baseline methods, which is attributed to its ability to explore the relationship between molecular substructures of various scales and side effects. Nevertheless, the SA-DDI decoder fails to consider the impact of various side effects, which results in suboptimal performance.

To address these challenges, CASE integrates biochemical comprehensive information about drugs, employs a fine-grained drug modeling technique, and dynamically extracts substructures of varying sizes while assessing the impact of excluded substructures on the side effects of numerous drugs. Furthermore, during the decoding process, CASE extensively investigates the potential correlations among side effects, resulting in excellent performance in predicting polypharmacy side effects.

5.3.2. *Unbalanced data set*

In 5.3.1, we adopt the same strategy proposed by Wang et al. [48] to evaluate the overall performance of our method. This strategy involves constructing a negative sample for each positive sample to form a balanced test set. However, our objective is to delve deeper into the performance of CASE in a realistic scenario and comprehensively evaluate the capabilities of this method. To achieve this, we conduct experiments using different negative and positive ratios. Specifically, we adjust the method of constructing negative samples to generate either 2

Table 5
The results of unbalanced data sets with different proportions.

	ACC	Prec	Rec	F1	AUROC	AUPRC
CASE (1 : 1)	88.70 ± 0.04	83.62 ± 0.03	96.45 ± 0.10	89.56 ± 0.04	94.75 ± 0.05	91.16 ± 0.07
CASE (1 : 2)	84.38 ± 0.07	72.63 ± 0.04	83.70 ± 0.09	78.33 ± 0.06	90.45 ± 0.07	82.18 ± 0.11
CASE (1 : 3)	82.93 ± 0.08	65.66 ± 0.03	66.48 ± 0.08	66.07 ± 0.07	90.11 ± 0.06	74.10 ± 0.09

or 3 negative samples for each positive sample. Subsequently, we segregate the training set and the test set accordingly.

As suggested in Table 5, it is evident that increasing the proportion of negative samples has resulted in a slight decrease in all performance indicators of the CASE model. However, it is important to note that the model's performance still remains satisfactory. A closer analysis reveals significant declines in indicators such as Prec, Rec, F1 score and AUPRC. The decline in Prec and Rec can be attributed to the sensitivity of these metrics to class imbalance issues, which can affect the model's ability to identify positive categories accurately. Prec represents the proportion of positive samples predicted by the model, while Rec represents the proportion of positive samples successfully identified by the model out of the total positive samples. With an increase in the number of negative samples, the model tends to predict more negative samples, posing a greater challenge in capturing a few positive categories in the test set. This leads to a decline in Prec and Rec. Furthermore, adjusting the sample ratio may result in the model not fully learning certain categories or positive features, further contributing to the decline in Prec and Rec on the test set. The F1 score offers a comprehensive assessment of Precision (Prec) and Recall (Rec), serving as a reflection of the model's overall performance in predicting positive cases. On the other hand, AUPRC represents the area under the Precision-Recall curve, with Precision (Prec) and Recall (Rec) as the pivotal metrics for its calculation. Thus, as Precision (Prec) and Recall (Rec) decrease, the scores of AUPRC and F1 also decrease.

5.3.3. Ablation experiment

To verify the effectiveness of the MS-based view, the BI-based view, the drug representation contrastive learning and the criterion-guided decoder, we conduct ablation experiment on the following CASE variants.

- CASE_w/o MS: The MS-based view is removed and the encoder learns drug embeddings only from the BI-based view.
- CASE_w/o BI: The BI-based view is removed and the encoder learns drug embeddings only from the MS-based view.
- CASE_w/o CL: Instead of using contrastive learning to refine drug embedding, the representations of drugs in two views are concatenated and fed into the criterion-guided encoder.
- CASE_w/o HL: Using MLP as the decoder and replacing the hinge loss with the cross-entropy loss.

Besides, we further introduce two interactive ablation variants to explore the joint effects of input modality and decoding strategy:

- CASE_w/o MS+HL: Both the MS view and the criterion-guided decoder are removed. The model relies solely on the BI view and uses a standard MLP decoder with cross-entropy loss.
- CASE_w/o BI+HL: Both the BI view and the criterion-guided decoder are removed. The model relies solely on the MS view with the same simplified decoding strategy.

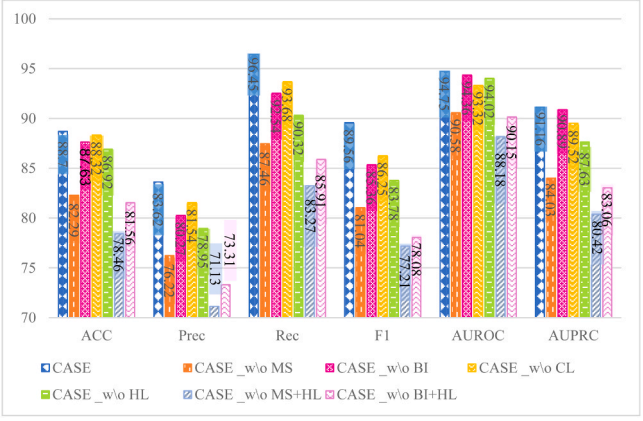


Fig. 7. Performance of CASE to other variants at all evaluation metrics.

Fig. 7 demonstrates the superior performance of the CASE model over the other four variants across all metrics. Specifically, the CASE_w/o MS exhibits the weakest performance compared to the variants that include structural information on drug substructures. This can be attributed to its sole reliance on a single piece of drug biochemical information as the final representation for predicting DDI characteristics. These results underscore the significance of integrating information from the drug substructure to achieve an accurate representation, and the omission of this information severely impairs the model's predictive performance. Furthermore, the CASE_w/o BI, which does not utilize drug similarity, drug target association, and drug SMILES information, obtains an insufficiently comprehensive drug representation, resulting in inferior performance compared to the CASE model. The lower performance observed for the CASE_w/o HL compared to the CASE model further supports the necessity of a well-designed decoder and the effectiveness of employing hinge loss. Finally, we replace contrastive learning with concatenating the representations of drugs in two views, resulting in the CASE_w/o CL exhibiting inferior performance. In addition to the single-module variants, Fig. 7 also reports the performance of the two interactive ablation variants, i.e., CASE_w/o MS+HL and CASE_w/o BI+HL, where both an input modality and the criterion-guided decoder are removed simultaneously. Compared to their respective single-module counterparts, these interactive variants show a more substantial decline in performance across all evaluation metrics. For example, CASE_w/o MS+HL significantly underperforms both CASE_w/o MS and CASE_w/o HL, suggesting that the absence of structural information cannot be sufficiently compensated by the remaining components, especially under a simplified decoding strategy. Similarly, CASE_w/o BI+HL exhibits a marked drop in performance relative to CASE_w/o BI and CASE_w/o HL, further confirming that integrating biochemical features and employing a decoding mechanism tailored to side effect semantics are both essential for reliable DDI prediction.

In a word, our finding reveals that the CASE model consistently outperforms all other variants, emphasizing the importance of integrating both drug molecular structure information and biochemical information simultaneously. The use of contrastive learning further enhances the accuracy of drug representations. Moreover, the application of criterion-guided decoders to analyse the relationships between side effects significantly improves the model's predictive effectiveness.

5.4. Parameter sensitivity

In this subsection, we focus on investigating the influence of specific hyperparameters within CASE. The three critical hyperparameters under consideration are denoted as follows: γ, which governs the two boundary ratios; δ, the weight assigned to negative entries; ε, a parameter used to quantify the contrastive loss. To ensure a thorough analysis, each hyperparameter is individually examined while keeping the remaining hyperparameters fixed. The ensuing sections present the experimental results, visually depicted in Fig. 8.

(1) Effect of the hyperparameter γ that controls the two boundary ratios.

The outcomes pertaining to the hyperparameter γ are prominently presented in Fig. 8. It is evident from the results that the model achieves optimal performance across all evaluation metrics when γ is set to 0.5. However, deviations from this optimal value, lead to a discernible decline in the model's predictive capacity. We propose several plausible reasons to explain these observations. When γ assumes values less than 0.5, the proximity of the upper and lower bounds becomes excessively narrow. Consequently, the model fails to effectively discern the accurate

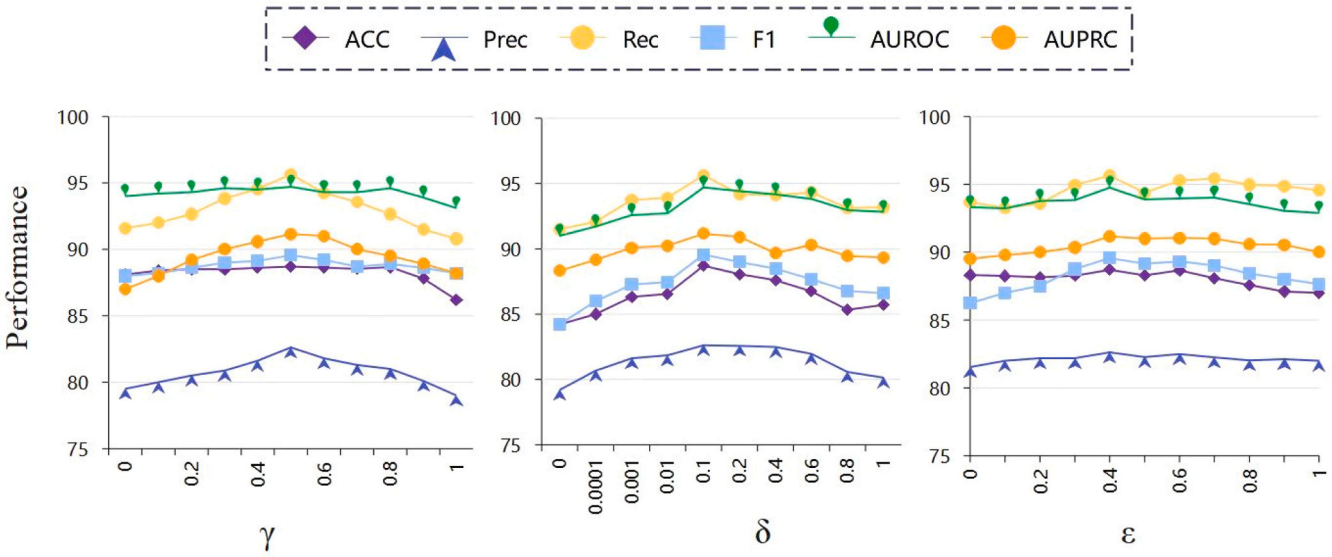


Fig. 8. Hyperparameter sensitivity analysis of CASE.

upper and lower boundaries, adversely impacting its ability to make precise predictions. This close confinement hinders the model from adequately capturing the intricacies of the underlying data, consequently hampering its predictive accuracy. Conversely, when γ exceeds 0.5, a significant widening of the upper and lower bounds occurs. This limits the comprehensive exploration of heterogeneous information encoded within distinct drug pairs, leading to a considerable reduction in its predictive performance.

(2) Effect of the weight δ of the negative entries.

Fig. 8 shows that as δ varies within the range of 0–1, the CASE's performance initially experiences improvement, followed by a subsequent decline. The optimal performance is achieved when δ sets a value of 0.1. Our analysis reveals that the augmentation of δ allows the CASE model to assimilate more informative content from negative entries, thus enhancing its discriminative capabilities. However, this gain in useful information comes at the expense of introducing additional noise during the model learning process. Consequently, the model's performance experiences a deterioration beyond an optimal δ value, which indicates an increased influence of the introduced noise.

(3) Effect of the weight ε of contrastive loss.

The hyperparameter ε is subjected to a variation across $\{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$, while keeping γ fixed at 0.5 and δ fixed at 0.1 throughout the experiments. The results, presented in Fig. 8, illustrate the CASE's performance across the range of ε values. Remarkably, the proposed method demonstrates stability over the spectrum of ε values. As Fig. 8 illustrates, the optimal performance is achieved when ε assumes a value of 0.4 across all evaluation metrics. This suggests that a moderate emphasis on contrastive supervision provides the best balance between preserving view-specific information and enforcing cross-view consistency. Setting ε too small may result in inadequate cross-view alignment between the molecular structure and biochemical information views. Conversely, an overly large ε may overemphasize the contrastive objective, thereby diminishing the contribution of the primary task supervision during optimization.

5.5. Training efficiency of CASE

To showcase the effectiveness of CASE, we present a visual comparison of the training process curves. Fig. 9. illustrates the training loss

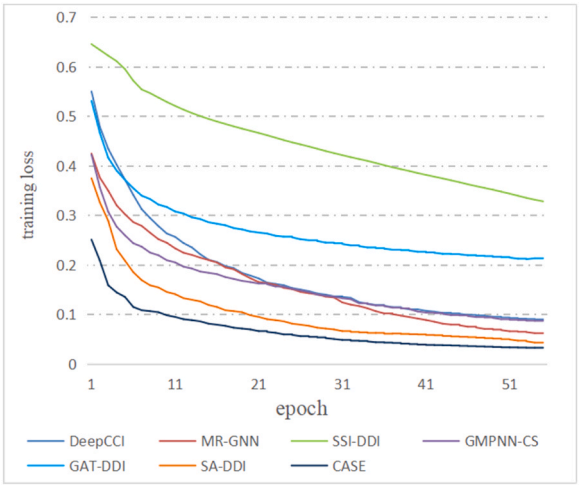


Fig. 9. With the increased epoch, the training loss of CASE and baseline models changes, in which CASE and its baselines are represented by different colors.

of CASE and the baselines, distinguished by various colors. From the visualization, it is evident that CASE exhibits the fastest convergence speed and maintains a stable training process. This can be attributed to several factors. Firstly, CASE employs directed message passing to extract rich and valuable features from the molecular graph data, leading to improved understanding of the drug's structural and chemical properties and obtaining higher-quality molecular graph representations. Additionally, CASE incorporates advanced atom-centered attention and substructure-aware pooling, improving the efficiency and accuracy of information transfer and allowing for better capture of key features in molecular graphs during training. This facilitates the convergence of the model. Lastly, integrating the prediction of various side effects into low-rank matrix learning significantly enhances model efficiency.

In addition to the loss curves shown in Fig. 9, we report both the training and inference efficiency of CASE compared with baseline models on a single NVIDIA A100 GPU. During training, CASE requires approximately 2.73 s per batch, slightly higher than SA-DDI (1.8 s per batch) due to its more complex architecture, including dual-view encoding, contrastive learning, and a criterion-guided decoder.

Table 6
Top 9 drugs predicted by CASE that are most likely to have side effects with Aliskiren.

Drug name	Side effect	Evidence
Pyrimethamine	Sarcoma	PMID: 25943187 [52]
Tioconazole	Breast inflammation	PMID: 23121378 [51]
Carvedilol	Abnormal ECG	drugs.com
Quinapril	Haematochezia	unconfirmed
Valsartan	Micturition urgency	PMID: 17658393 [53]
Carvedilol	Joint swelling	drugs.com
Prednisone	High blood pressure	drugs.com
Ipratropium bromide	Paraesthesia	unconfirmed
Ramipril	Pain in throat	PMID: 18205098 [54]

Nevertheless, the training time remains practical for large-scale DDI datasets. For inference, the average latency per sample is 0.29 ms for CASE versus 0.25 ms for SA-DDI, indicating that the proposed model maintains low and comparable prediction time to the baseline despite its increased architectural complexity.

5.6. Case study: Aliskiren

In this subsection, we perform a case study to demonstrate the robust performance of CASE. We analyze the prediction results meticulously and concentrate on the top 9 drug pairs involving Aliskiren, as shown in Table 6. For these 9 pairs of drugs, we apply Drugs.com, and PubMed tools to find evidence supporting them. From Table 6, we can observe that there is evidence for 7 of the 9 DDI pairs, which further indicates the validity of CASE. For example, CASE predicts that when patients take tioconazole and Aliskiren at the same time, they may suffer from breast inflammation, which is proved by the literature [51].

It is noteworthy that CASE's predictions indicate that Aliskiren and Carvedilol may individually induce distinct side effects, namely abnormal ECG and joint swelling, respectively. These predictions align with evidence documented on Drug.com, further validating the accuracy of our model. More importantly, CASE provides actionable insights for clinical decision-making. For instance, the predicted interaction between Aliskiren and Tioconazole—associated with breast inflammation—may prompt caution in patients with hormone-sensitive conditions. Such predictions go beyond simple detection by offering side-effect-specific guidance that can influence prescribing choices, particularly for vulnerable populations like the elderly or those with comorbidities. In addition, CASE produces probabilistic estimations for each drug pair–side effect association, rather than binary outputs. This enables clinicians to assess the likelihood of adverse events, prioritize patient monitoring, and explore safer alternatives. This highlights the superior performance of our model in predicting polypharmacy-induced side effects compared to earlier models. Prior approaches have typically concentrated on forecasting the occurrence of individual side effects resulting from drug interactions, or simply detecting their presence without providing detailed specific probability predictions.

In addition to the predictions that are supported by existing literature, some predicted side effects currently lack direct clinical confirmation. For example, CASE predicts that the combination of quinapril and aliskiren may result in haematochezia, which is not yet documented in current databases. However, quinapril has been reported to occasionally cause abdominal discomfort or bleeding, while aliskiren may induce mucosal irritation. As dual renin–angiotensin system inhibitors, their combined use could potentially exacerbate vascular fragility or gastrointestinal mucosal stress, thereby increasing the risk of bleeding, even though this specific event remains underreported. Similarly, the predicted paraesthesia arising from the co-administration of aliskiren and ipratropium bromide has not been clinically validated. Nevertheless, both drugs can affect the autonomic nervous system, and their interaction may plausibly lead to peripheral neurological symptoms such as tingling or numbness.

Importantly, the absence of published evidence does not necessarily invalidate these predictions. Instead, it may reflect limitations in current clinical observations or underreporting within pharmacovigilance databases. Such unconfirmed predictions may represent rare, latent, or previously overlooked adverse events. Therefore, CASE's ability to flag these interactions highlights its potential as a discovery tool for emerging drug safety risks, offering early warnings that can inform future experimental validation or population-based studies.

From a clinical understanding and decision-making perspective, this interpretability is particularly valuable. Not only does it identify drug interactions supported by existing evidence, but it also surfaces plausible yet underrecognized risks, contributing meaningful insights that can assist in patient safety assessment, inform drug combination guidelines, and support personalized therapeutic strategies in complex polypharmacy scenarios.

5.7. Case study: exploration of CASE's drug embeddings

To offer a more intuitive understanding of the relationships between drug representations, we employ t-SNE [55] to visualize drug embedding vectors in Fig. 10. t-SNE is a non-linear dimensionality reduction technique that maps high-dimensional vectors into a two-dimensional space, facilitating visualization. We rank the drug pairs predicted by the model by the number of side effects and differentiate them in the visualization: node pairs with the highest number of side effect relationships are marked in red, those with some side effect relationships are in yellow, and those without side effect relationships are in green. The node pairs highlighted in yellow indicate the existence of specific side effects associated with the drug pair, with a minimum count of 1. It is important to note that this count is lower than the number of drug pairs highlighted in red, which represent stronger associations with multiple side effects.

The close proximity of the red-labeled drug pairs "loxapine" and "entacapone," "thymidine" and "bortezomib," and "pravastatin" and "calcium" in the embedding space indicates a high degree of similarity in their representations. Concurrently, CASE predicts a broader spectrum of potential side effects among these pairs, aligning with chemical knowledge. Specifically, the imidazole ring structures of entacapone and loxapine may engage in stacking or other interactions, potentially affecting their in vivo metabolism, distribution, and potency [56]. Furthermore, the borate moiety of bortezomib may interact with the purine ring structure of thymidine, which could interfere with

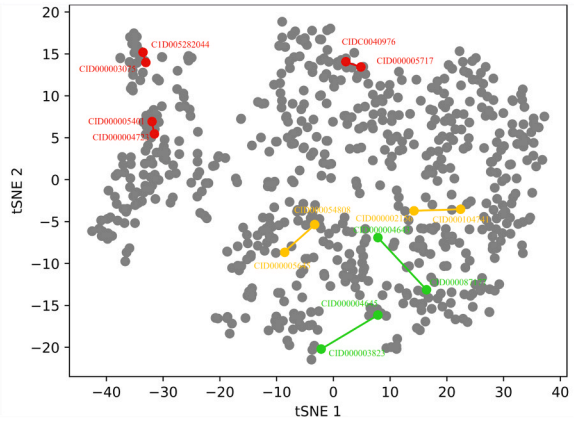


Fig. 10. t-SNE visualization of drug embeddings learned by CASE. Each dot represents a drug, color-coded by the number of predicted polypharmacy side effects: red indicates drug pairs with the highest number of predicted interactions, yellow denotes pairs with fewer but specific side effects, and green corresponds to pairs without any predicted side effects. Notably, red-labeled drug pairs cluster more closely, suggesting that CASE effectively captures semantic and pharmacological similarity in the embedding space.

thymidine's binding to adenosine receptors or alter its activity [57]. When pravastatin is combined with calcium, calcium ions may interact with the pravastatin target, interfering with the drug's binding or modulation, thereby influencing its efficacy and side effects [58]. In contrast, the green-labelled drug pairs "nevirapine" and "epinephrine," and "thymine" and "pantoprazole," exhibit the greatest distance from each other in the examples provided, with the model indicating no predicted side effects between them.

The visualization results support our hypothesis that the greater the similarity between embedded representations, the higher the likelihood of potential side effects occurring between drugs. In summary, CASE not only demonstrates the ability to derive precise drug representations but also makes reliable predictions.

6. Conclusion

In this paper, we propose a novel framework CASE for polypharmacy drug side effect prediction. CASE incorporates an adaptive substructure encoder and a biochemical comprehensive feature aggregator, captures the microscopic (drug molecular diagram) features of drugs through directed message passing with subgraph attention, and integrates the biochemical knowledge of drugs with multi-layer perceptron to master the macroscopic features of drugs. It fuses multi-scale drug information using a contrastive learning mechanism and further enhances the discriminative power of drug representations. In addition, CASE uses criterion-guided decoders to introduce multiple boundaries and utilizes hinge loss to guide model learning. By modeling side effect dependencies and incorporating interpretable biochemical knowledge, CASE not only improves predictive accuracy but also enhances clinical interpretability. This enables better identification of high-risk drug combinations and provides valuable decision support in multi-drug prescription scenarios. Experimental results demonstrate that CASE outperforms both traditional and state-of-the-art (SOTA) methods. Specifically, CASE achieves the best performance with 88.70 % accuracy, 83.62 % precision, 96.45 % recall, 89.56 % F1-score, 94.75 % AUROC, and 91.16 % AUPRC. Compared to the best baseline method (SA-DDI), CASE improves AUROC and AUPRC by 1.58 % and 0.65 %, respectively. Ablation studies further verify the effectiveness of each component, including the MS-based view, BI-based view, contrastive learning, and criterion-guided decoder. Moreover, we evaluated CASE under different negative-positive sample ratios (1:1, 1:2, 1:3), and CASE maintained competitive performance even under increasingly imbalanced settings, showing strong generalizability.

CASE is designed not only to enhance predictive accuracy, but also to generate clinically meaningful predictions that can inform pharmacological understanding and real-world medical decisions. Some of CASE's predictions have found supporting evidence in existing literature, thereby reinforcing their clinical interpretability and potential utility in decision-making. These validated cases demonstrate CASE's ability to generate pharmacologically meaningful insights by integrating molecular structures with biochemical properties. Meanwhile, although certain predictions have not yet been validated in current literature, such predictions may serve as early warning signals for rare or underreported adverse effects, guiding future clinical research and pharmacovigilance studies.

Overall, these results suggest that CASE is not only effective in predictive accuracy but also robust across different data scenarios. Its dual-view encoding and decoder design enable interpretable, scalable, and clinically valuable modeling of polypharmacy risk—offering a promising tool to support safe medication decisions. As part of our future direction, we aim to integrate more curated datasets with high-confidence annotations and explore label denoising techniques to enhance the model's resilience to label noise.

CRediT authorship contribution statement

Huifang Ma: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yike Wang:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhixin Li:** Writing – review & editing, Validation, Supervision, Conceptualization. **Zihao Gao:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Liang Chang:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Code availability

The implementation of CASE is available at https://github.com/222249yk/CASE.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] N. Masnoon, S. Shakib, L. Kalisch-Ellett, G.E. Caughey, What is polypharmacy? A systematic review of definitions, BMC Geriatr. 17 (2017) 1–10.
[2] R.W. Hoel, R.M.G. Connolly, P.Y. Takahashi, Polypharmacy management in older patients. in: Mayo Clinic Proceedings, Elsevier, 2021, pp. 242–256.
[3] S. Subha, M. Shanmugathai, A. Prasanth, S.S. Varagi, V. Dhanashree, Digital Transformation in the Pharmaceutical and Biotech Industry: Challenges and Research Directions, Digital Twins in Industrial Production Smart Manufacturing: An Understanding of Principles, Enhancers, Obstacles, (2024) 297-324.
[4] A. Bharathi, U. Hemamalini, A. Prasanth, Augmented machine learning towards smart self-powered sensing systems. in: Self-Powered Sensors, Elsevier, 2025, pp. 31–45.
[5] S. Radhika, A. Prasanth, K.D.J.B.S.P. Sowndarya, Control, A Reliable speech emotion recognition framework for multi-regional languages using optimized light gradient boosting machine classifier, 105 (2025) 107636.
[6] M. Zitnik, M. Agrawal, J. Leskovec, Modeling polypharmacy side effects with graph convolutional networks, Bioinformatics 34 (2018) i457–i466.
[7] Y. Wang, H. Ma, R. Zhang, Z. Gao, Drug side effects prediction via heterogeneous Multi-Relational graph convolutional networks, in: 2022 IEEE 34th International Conference on Tools with Artificial Intelligence (ICTAI), IEEE, 2022, pp. 1093–1097.
[8] Y. Chen, T. Ma, X. Yang, J. Wang, B. Song, X. Zeng, MUFFIN: multi-scale feature fusion for drug–drug interaction prediction, Bioinformatics 37 (2021) 2651–2658.
[9] Y. Wang, Y. Min, X. Chen, J. Wu, Multi-view graph contrastive representation learning for drug-drug interaction prediction, Proc. web Conf. 2021 (2021) 2921–2933.
[10] Y. Ding, J. Tang, F. Guo, Identification of drug-side effect association via semisupervised model and multiple kernel learning, IEEE J. Biomed. Health Inform. 23 (2018) 2619–2632.
[11] Y. Ding, J. Tang, F. Guo, Identification of drug-side effect association via multiple information integration with centered kernel alignment, Neurocomputing 325 (2019) 211–224.
[12] D. Galeano, S. Li, M. Gerstein, A. Paccanaro, Predicting the frequencies of drug side effects, Nat. Commun. 11 (2020) 4575.
[13] J. Wang, Y. Xiao, X. Shang, J. Peng, Predicting drug–target binding affinity with cross-scale graph contrastive learning, Brief. Bioinforma. 25 (2024) bbad516.
[14] Z. Tian, Y. Yu, F. Ni, Q. Zou, Drug-target interaction prediction with collaborative contrastive learning and adaptive self-paced sampling strategy, BMC Biol. 22 (2024) 216.

[15] A. Deac, Y.-H. Huang, P. Veličković, P. Liò, J. Tang, Drug-Drug adverse effect prediction with graph Co-Attention, stat 1050 (2019) 2.
[16] A.K. Nyamabo, H. Yu, Z. Liu, J.-Y. Shi, Drug–drug interaction prediction with learnable size-adaptive molecular substructures, Brief. Bioinforma. 23 (2022) bbab441.
[17] A.K. Nyamabo, H. Yu, J.-Y. Shi, SSI–DDI: substructure–substructure interactions for drug–drug interaction prediction, Brief. Bioinforma. 22 (2021) bbab133.
[18] Z. Li, S. Zhu, B. Shao, X. Zeng, T. Wang, T.-Y. Liu, DSN-DDI: an accurate and generalized framework for drug–drug interaction prediction by dual-view representation learning, Brief. Bioinforma. 24 (2023) bbac597.
[19] K. Huang, C. Xiao, T. Hoang, L. Glass, J. Sun, Caster: predicting drug interactions with chemical substructure representation, Proc. AAAI Conf. Artif. Intell. (2020) 702–709.
[20] Z. Yang, W. Zhong, Q. Lv, C.Y.-C. Chen, Learning size-adaptive molecular substructures for explainable drug–drug interaction prediction by substructure-aware graph neural network, Chem. Sci. 13 (2022) 8693–8703.
[21] Z. Tang, G. Chen, H. Yang, W. Zhong, C.Y.-C. Chen, L. Systems, DSIL-DDI: a domain-invariant substructure interaction learning for generalizable drug–drug interaction prediction, IEEE Trans. Neural Netw. (2023).
[22] D. Niu, L. Xu, S. Pan, L. Xia, Z. Li, SRR-DDI: a drug–drug interaction prediction model with substructure refined representation learning based on self-attention mechanism, Knowl. Based Syst. 285 (2024) 111337.
[23] S. Chen, I. Semenov, F. Zhang, Y. Yang, J. Geng, X. Feng, Q. Meng, K. Lei, An effective framework for predicting drug–drug interactions based on molecular substructures and knowledge graph neural network, Comput. Biol. Med. 169 (2024) 107900.
[24] Y. Zhong, G. Li, J. Yang, H. Zheng, Y. Yu, J. Zhang, H. Luo, B. Wang, Z. Weng, Learning motif-based graphs for drug–drug interaction prediction via local–global self-attention, Nat. Mach. Intell. 6 (2024) 1094–1105.
[25] J. Sun, H. Zheng, HDN-DDI: a novel framework for predicting drug-drug interactions using hierarchical molecular graphs and enhanced dual-view representation learning, BMC Bioinforma. 26 (2025) 28.
[26] S. Vilar, R. Harpaz, E. Uriarte, L. Santana, R. Rabadan, C. Friedman, Drug—drug interaction through molecular structure similarity analysis, J. Am. Med. Inform. Assoc. 19 (2012) 1066–1074.
[27] I. Abdelaziz, A. Fokoue, O. Hassanzadeh, P. Zhang, M. Sadoghi, Large-scale structural and textual similarity-based mining of knowledge graph to predict drug–drug interactions, J. Web Semant. 44 (2017) 104–117.
[28] T. Ma, C. Xiao, J. Zhou, F. Wang, Drug similarity integration through attentive Multi-view graph Auto-Encoders, Proc. TwentySeven. Int. Jt. Conf. Artif. Intell. (2018).
[29] Y. Deng, X. Xu, Y. Qiu, J. Xia, W. Zhang, S. Liu, A multimodal deep learning framework for predicting drug–drug interaction events, Bioinformatics 36 (2020) 4316–4322.
[30] Z. Yang, K. Tong, S. Jin, S. Wang, C. Yang, F. Jiang, CNN-Siam: multimodal siamese CNN-based deep learning approach for drug-drug interaction prediction, BMC Bioinforma. 24 (2023) 110.
[31] R. Masumshah, C. Eslahchi, DPSP: a multimodal deep learning framework for polypharmacy side effects prediction, Bioinforma. Adv. 3 (2023) vbad110.
[32] M. Jiang, G. Liu, Y. Su, W. Jin, B. Zhao, Hierarchical multi-relational graph representation learning for large-scale prediction of drug-drug interactions, IEEE Trans. Big Data (2025).
[33] M. Asfand-E-Yar, Q. Hashir, A.A. Shah, H.A.M. Malik, A. Alourani, W. Khalil, Multimodal CNN-DDI: using multimodal CNN for drug to drug interaction associated events, Sci. Rep. 14 (2024) 4076.
[34] D. Kumari, D. Agrawal, A. Nema, N. Raj, S. Panda, J. Christopher, J.K. Singh, S. Behera, A study on improving drug–drug interactions prediction using convolutional neural networks, Appl. Soft Comput. 166 (2024) 112242.
[35] R. Li, Y. Zhang, H. Sun, S. Lin, G. Jia, Y. Fang, C. Zhang, X. Song, J. Zhao, L. Hu, Towards interpretable drug interaction prediction via dual-stage attention and Bayesian calibration with active learning, PeerJ Comput. Sci. 11 (2025) e2847.
[36] M.W. Harrold, R.M. Zavod, Basic concepts in medicinal chemistry, Drug Dev. Ind. Pharm. (2014).
[37] J. Wei, L. Lu, T. Shen, Predicting drug–protein interactions by preserving the graph information of multi source data, BMC Bioinforma. 25 (2024) 10.
[38] M. Kuhn, C. von Mering, M. Campillos, L.J. Jensen, P. Bork, STITCH: interaction networks of chemicals and proteins, Nucleic Acids Res. 36 (2007) D684–D688.
[39] J.J. Irwin, B.K. Shoichet, ZINC– a free database of commercially available compounds for virtual screening, J. Chem. Inf. Model. 45 (2005) 177–182.
[40] P. Bachman, R.D. Hjelm, W. Buchwalter, Learning representations by maximizing mutual information across views, Adv. Neural Inf. Process. Syst. (2019) 32.
[41] N.P. Tatonetti, P.P. Ye, R. Daneshjou, R.B. Altman, Data-driven prediction of drug effects and interactions, Sci. Transl. Med. 4 (2012) 125ra131.
[42] C. Knox, M. Wilson, C.M. Klinger, M. Franklin, E. Oler, A. Wilson, A. Pon, J. Cox, N. E. Chin, S.A. Strawbridge, DrugBank 6.0: the DrugBank knowledgebase for 2024, Nucleic Acids Res. 52 (2024) D1265–D1275.
[43] P. Velickovic, W. Fedus, W.L. Hamilton, P. Liò, Y. Bengio, R.D. Hjelm, Deep graph infomax, ICLR 2 (2019) 4.
[44] S. Kwon, S. Yoon, End-to-end representation learning for chemical-chemical interaction prediction, IEEE/ACM Trans. Comput. Biol. Bioinforma. 16 (2018) 1436–1447.
[45] N. Xu, P. Wang, L. Chen, J. Tao, J. Zhao, Mr-gnn: Multi-resolution and dual graph neural network for predicting structured entity interactions, arXiv preprint arXiv: .09558, (2019).
[46] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, Y. Bengio, Graph attention networks, arXiv preprint arXiv:.10903, (2017).
[47] T. Lyu, J. Gao, L. Tian, Z. Li, P. Zhang, J. Zhang, MDNN: a multimodal deep neural network for predicting Drug-Drug interaction events, Ijcai (2021) 3536–3542.
[48] Z. Wang, J. Zhang, J. Feng, Z. Chen, Knowledge graph embedding by translating on hyperplanes, Proc. AAAI Conf. Artif. Intell. (2014).
[49] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, Pytorch: an imperative style, high-performance deep learning library, Adv. Neural Inf. Process. Syst. (2019) 32.
[50] J.Y. Ryu, H.U. Kim, S.Y. Lee, Deep learning improves prediction of drug–drug and drug–food interactions, Proc. Natl. Acad. Sci. 115 (2018) E4304–E4311.
[51] H.-H. Parving, B.M. Brenner, J.J. McMurray, D. De Zeeuw, S.M. Haffner, S. D. Solomon, N. Chaturvedi, F. Persson, A.S. Desai, M. Nicolaides, Cardiorenal end points in a trial of aliskiren for type 2 diabetes, N. Engl. J. Med. 367 (2012) 2204–2213.
[52] T.B. Stage, K. Brøsen, M.M.H. Christensen, A comprehensive review of drug–drug interactions with metformin, Clin. Pharmacokinet. 54 (2015) 811–824.
[53] J. Shen, W. Feng, Y. Wang, Q. Zhao, B.L. Flavorta, J. Lu, Efficacy and safety of aliskiren combination therapy: a protocol for an umbrella review, BMJ Open 11 (2021) e043807.
[54] Y. Uresin, A.A. Taylor, C. Kilo, D. Tschöpe, M. Santonastaso, G. Ibram, H. Fang, A. Satlin, Efficacy and safety of the direct renin inhibitor aliskiren and ramipril alone or in combination in patients with diabetes and hypertension, J. ReninAngiotensinAldosterone Syst. 8 (2007) 190–200.
[55] L. Van der Maaten, G. Hinton, Visualizing data using t-SNE, J. Mach. Learn. Res. 9 (2008).
[56] P. Sharma, C. LaRosa, J. Antwi, R. Govindarajan, K.A. Werbovetz, Imidazoles as potential anticancer agents: an update on recent studies, Molecules 26 (2021) 4213.
[57] J. Adams, M. Kauffman, Development of the proteasome inhibitor Velcade™(Bortezomib), Cancer Investig. 22 (2004) 304–311.
[58] C. Stancu, A. Sima, Statins: mechanism of action and effects, J. Cell. Mol. Med. 5 (2001) 378–387.